CAPSTONE PROJECT REPORT

(Project Term January-May 2025)

Dynamic Stock Price Forecasting using Machine Learning and Real-Time Data Integration

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CSERGC0295

CSE439

Under the Guidance of

Dr. Parampreet Kaur

Associate Professor

School of Computer Science and Engineering

PAC Form

TOPIC APPROVAL PERFORMA

School of Computer Science and Engineering (SCSE)

Program : P132::B.Tech. (Computer Science and Engineering)

COURSE CODE : CSE439 **REGULAR/BACKLOG :** Regular **GROUP NUMBER :** CSERGC0295

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Qualification : PHD-CSE **Research Experience :** 12 yrs.

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SPECIALIZATION AREA : Software Testing **Supervisor Signature:** 18758

PROPOSED TOPIC : Dynamic Stock Price Forecasting using Machine Learning and Real-Time Data Integration

Qualitative Assessment of Proposed Topic by PAC		
Sr.No.	**Parameter**	**Rating (out of 10)**
1	Project Novelty: Potential of the project to create new knowledge	7.28
2	Project Feasibility: Project can be timely carried out in-house with low-cost and available resources in the University by the students.	6.92
3	Project Academic Inputs: Project topic is relevant and makes extensive use of academic inputs in UG program and serves as a culminating effort for core study area of the degree program.	6.92
4	Project Supervision: Project supervisor's is technically competent to guide students, resolve any issues, and impart necessary skills.	8.36
5	Social Applicability: Project work intends to solve a practical problem.	7.28
6	Future Scope: Project has potential to become basis of future research work, publication or patent.	6.72

PAC Committee Members		
PAC Member (HOD/Chairperson) Name: Dr. Dalwinder Singh	UID: 11265	Recommended (Y/N): Yes
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Final Topic Approved by PAC: **Dynamic Stock Price Forecasting using Machine Learning and Real-Time Data Integration**

Overall Remarks: Approved

PAC CHAIRPERSON Name: 14537::Dr. Rekha **Approval Date:** 03 Mar 2025

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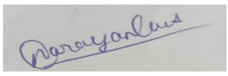
DECLARATION

We hereby declare that the project work "Dynamic Stock Price Forecasting using Machine Learning and Real-Time Data Integration" is an authentic record of our own work carried out as requirements of Capstone Project for the award of B. Tech degree in Computer Science and Engineering from Lovely Professional University, Phagwara under the guidance of Dr. Parampreet Kaur during January to June 2025. All the information furnished in this capstone project report is based on our own intensive work and is genuine.

Project Group Number: CSERGC0295

Narayan Gupta — Signature 
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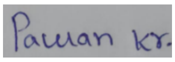
Pawan Kumar — Signature: 
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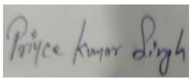
Prince Kumar Singh — Signature: 
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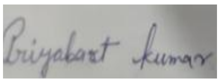
Priyabart Kumar — Signature: 
12110966 — Date:01/05/2025

CERTIFICATE

This is to certify that the declaration statement made by this group of students is correct to the best of my knowledge and belief. They have completed this Capstone Project under my guidance and supervision. The present work is the result of their original investigation, effort and study. No part of the work has ever been submitted for any other degree at any University. The Capstone Project is fit for the submission and partial fulfilment of the conditions for the award of B. Tech degree in Computer Science and Engineering from Lovely Professional University, Phagwara.

Dr. Parampreet Kaur

Associate Professor

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Date:

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We humbly take this opportunity to present our votes of thanks to all those guideposts who really acted as lightening pillars to enlighten our way throughout this project that has led to successful and satisfactory completion of this study. We are grateful to our mentor **Dr. Parampreet Kaur** for providing us with an opportunity to undertake this project at this university and providing us with all the bright and innovative ideas for making our project worthwhile running in an organization. We are highly thankful for his active support, valuable time and advice, whole-hearted guidance, sincere cooperation, and pain- taking involvement during the study and in completing the capstone project within the time stipulated. We are thankful to all those, particularly our friends, who have been instrumental in creating a proper, healthy and conductive environment and including new and fresh innovative ideas for us during the project. Without their help, it would have been extremelydifficult for us to prepare the project in a time-bound framework.

Narayan Gupta

Prince Kumar Singh

Priyabart Kumar

Pawan Kumar

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1. INTRODUCTION

The prediction of stock market price is important for the potential investors who desire to make profit from investment to manage risk and decision-making. Nevertheless, the detection is difficult for financial markets are noisy and volatile and the movements of stock prices are non-linear. In contrast to previous works, this study concentrates on boosting stock price forecasting based on modern machine learning methodologies, thus providing a more reliable data-based way to address the complexity of market. The analysis in this paper is based on an extensive set of TCS stock prices, which span of time is 2011-2024. The dataset contains the daily trading information, opening and closing prices, traded volume, and technical indicators. These features are then used to train predictive models for capturing market patterns behaviour over time. The paper also compares four types of machine learning methods: Long Short-Term Memory (LSTM), Random Forest, Decision Tree, and Support Vector Regression (SVR) as the baseline methods to provide the most accurate prediction model.

The LSTM neural network gave the best predictive accuracy of 96% when compared to other models tested. The capacity for LSTM networks to learn sequential dependencies via the memory cell structure, may be used to great effectiveness in modelling time series data, such as long-term tendencies of stock prices. This is particularly well suited for financial time series prediction, given the relevance of historical data in the formation of future patterns. The Random Forest model also proved a high-performing one as was to be expected, with an accuracy of 92%. Random Forest forms an ensemble of decision trees and combines their outputs, and accordingly it is able to model more complex interdependencies in the data while minimizing over-fitting. It offers a middle ground between correctness and interpretability. The accuracy of the Decision Tree model is 86%. It does not have the same level of predictive accuracy as the LSTM and Random Forest models, but it’s simple tree-based approach makes it easier to interpret and explain how the model makes decisions based on the input features. This interpretability makes it valuable as a means to gain insights into the underlying dynamics behind stock movements. The poorest performances were obtained using Support Vector Regression (SVR) (accuracy = 77%). Nevertheless, SVR worked because stock prices could be projected by transferring the input data to higher dimensions, which may help to expose more complex non-linear patterns. It’s still a useful approach when working with non-linear relationships in financial data, but it’s less accurate than using more complex models like LSTM.

In order to improve the performance of all models, a set of preprocessing methods were tried. These involved normalization, feature engineering and incorporation of live data feeds. These methodologies improved the models and made them more useful and robust in real financial markets.

In all, this work builds a connection between abstract machine learning theory and concrete market practice. Through rigorous model comparisons and state-of-the-art data processing techniques, it provides a strong support for data driven decision making, which is crucial especially in volatile and unpredictable markets.

2. SCOPE OF THE STUDY

The key aim of this project is to enhance a dynamic, accurate real-time stock price prediction system using state-of-the-art machine learning and deep learning techniques. Predicting stock market is an extremely complicated and volatile problem driven by complex nonlinear relationships. The target of this study is to bridge the hole between theoretical patterns and practical applications, by modeling long time practical record data of Tata Consultancy solutions (TCS) for the period of 2010-2024. With the application of algorithms including LSTM, SVR, Random Forest, and Tree models, the project strives to improve the prediction accuracy and offer the investor with actionable intelligence. Preprocessing uses a number of standard statistics preprocessing techniques such as normalization, deleting outliers, function definition to sample some tidy data and model validness. Also, because the Yahoo Finance APIs are used to consolidate data in real-time, the bot can be calibrated to the current value of the market. using various evaluation parameters e.g. mean Absolute error (MAE), Root means square errors (RMSE) and R^2. The simplest is the LSTM, at 96% accuracy, the only development model currently. The ultimate goal is to show that Machine Learning algorithms can be used for forecasting economic announcements so that traders can make a decision based on that information. Besides, by comparing several models, the mission identifies strengths, weaknesses, and points for improvement, leading to continual developments of system gaining knowledge of programs in inventory market appraisal and economic era improvements.

3. EXISTING SYSTEM

3.1. INTRODUCTION

Machine Learning (ML) has significantly revolutionized how we forecast the stock market by performing better than traditional methods such as technical and fundamental analysis. New methods such as Long Short-Term Memory (LSTM), Support Vector Regression (SVR), Random Forests, and Decision trees have been developed to accommodate the complex and dynamic nature of financial markets. These ML models have the ability to process large amounts of data, discover hidden patterns, and enhance forecasting precision, performing better than traditional models that often overlook important market information. Perhaps one of the largest advantages of ML models is that they can utilize various types of data, such as global market interconnectivity, social media sentiment, and technical indicators such as the Relative Strength Index (RSI). Conventional systems were typically based on linear models and limited data whereas, nullmodels exploit large and diverse datasets to find hidden patterns and subtle co variations. The consequence is higher precision of prediction, as well as more flexible adaptation to the real-time market. The use of models like those described here constitutes a revolution of the decision maker's art for investors and financial institutions. ML enables more accurate market predictions and strategic insights into market trends, showing users how to trade or invest wisely even in extremely volatile markets. Today’s market participants can rely on predictive models that are constantly learning and growing increasingly sophisticated as they consume more data — rather than succumbing to their emotions and trading recklessly. Also, all machine learning models using like Random Forest are ensemble learning and average out many models’ predictions and come up with 1 prediction and that reduces bias/overfitting and enables to generalize better. Random Forests models have a natural resistance to overfitting, they do learn very well and they do generalize well because it is averaging many weak learners, that’s crucial in the random nature of the markets.

3.2. EXISTING SOFTWARE

Previous approaches to stock price prediction have favored traditional statistical models such as linear regression and ARIMA, and simple machine learning models such as Support Vector Regression (SVR) and Decision Trees. Although these approaches make some kind of prediction, they are poor at coping with the nonlinear, dynamic nature of financial markets. Moreover, traditional approaches are typically accompanied by significant manual feature engineering, with over-reliance upon human expertise to choose and prepare input variables, thereby risking bias and confining scalability. These models are poor at responding to sudden market shifts, irregular trading patterns, or simultaneous influencing factors. The recent breakthrough in deep learning, i.e., the introduction of Long Short-Term Memory (LSTM) networks, made automatic learning from sequential data possible and improved the capacity to comprehend simple market behavior. Our system takes advantage of these breakthroughs through the use of LSTM models combined with real-time data integration, thus allowing more accurate and timely forecasting of the stock price.

3.3. DFD FOR PRESENT SYSTEM

0-Level Data Flow Diagram

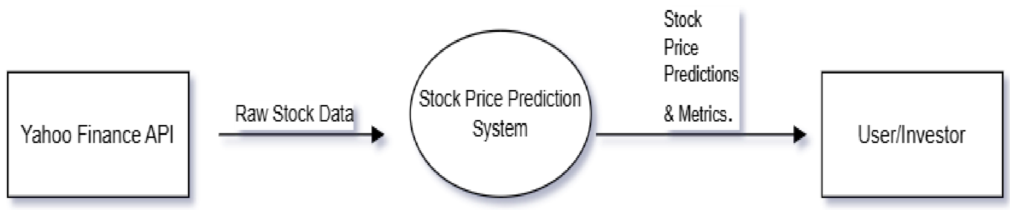

Fig. 1

1-Level Data Flow Diagram

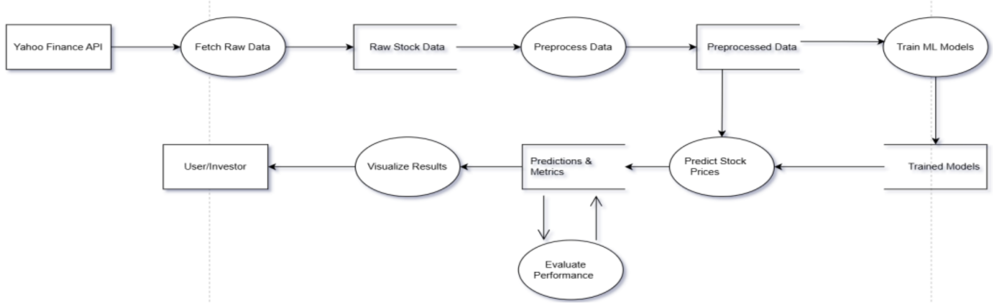

Fig. 2

2-Level DFD for Process: Preprocessed Data

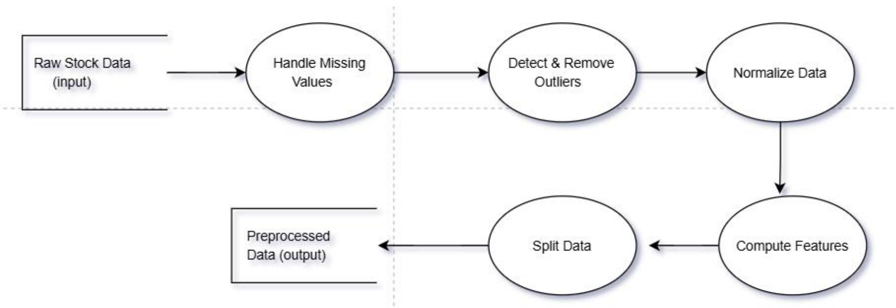

Fig. 3

2-Level DFD for Process: Train ML Models

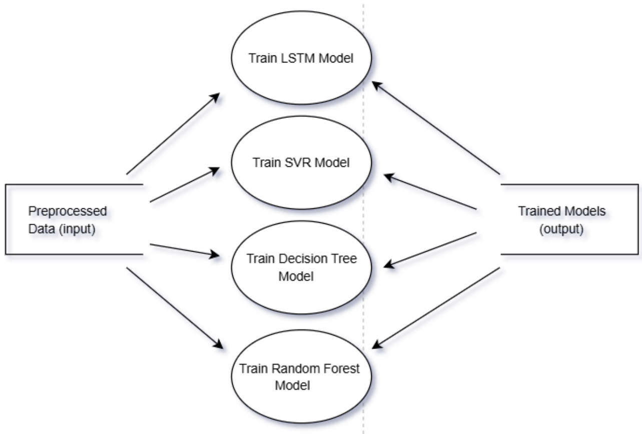

Fig. 4

Data Flow Diagrams (DFDs) representations the flow of the system are described on levels viz Level 0 for interaction with Yahoo Finance and the user; Level 1 which includes processes such as data fetching, preprocessing, training, prediction, Evaluation and visualization; Level 2 depiction about preprocessing (normalization, feature computation), training and prediction for each model; Level 3 for the training of LSTM (sequence preparation, hyperparameter tuning) in LSTM. The project plan focuses on a principled process from data retrieval to final visualization of results and will be designed to scale in capacity and accuracy. Functional requirements define the objectives and describe the data retrieval, preprocessing, the model training and the visualization, while the non-functional requirements are aimed at achieving an R^2 greater 0.90, the overall processing time less than 10 minutes and for the user interfaces to be intuitive. Testing involves functional tests for the verification of the assumptions, structural tests to validate the integrity of the code (e.g. GridSearchCV for SVR), multi-level testing (unit, integration, system, acceptance) and project-specific tests that approve that the LSTM is superior. The bibliography includes the references and any tools (in this case, scikit-learn and yfinance) that you use, to give credit where credit is due. In a nutshell, the system represents a stable, data-oriented answer for investors, utilizing state-of-the-art machine learning to make accurate predictions about stock prices.

4. PROBLEM ANALYSIS

4.1. PRODUCT DEFINITION

The output is a stock price prediction system based on machine learning algorithms like: LSTM, SVR, Decision Tree, and Random Forest for the prediction of TCS open Prices on 2010 – 2024 using the historical data of TCS. It retrieves, processes, and makes predictions using real-time financial market data off Yahoo Finance, With R^2, MAE, RMSE Metric and Visualizations to guide investors with their research and through their investments.

4.2. FEASIBILITY STUDY

Technological rationality and economic rationality of the proposed stock price forecast system are examined. Some other complex models such as Long Short-Term Memory (LSTM), Random Forest, Decision Tree, and Support Vector Regression (SVR) were also carefully assessed. LSTM performed better in terms of prediction accuracy (96%) confirming the suitability of this approach to capture complex dynamic patterns from financial time series. Economic viability is supported by the use of freely available data, open-source machine learning frameworks, and efficient algorithms that run on commodity hardware. Nonetheless, this method significantly lowers infrastructure and operational costs. The scalability and real-time capability of the system also make it attractive for real-world deployment. Acknowledged difficulties relating to high market volatility, inconsistent data points, and over-fitting algorithms are mitigated through ad-hoc strategies, including the re-training of models, fact manipulation and detection of anomalies. Overall, the feasibility study determines that the system is technically feasible, economically feasible, and commercially poised for deployment into financial markets.

Table 1: Performance Comparison of Machine Learning Models

Model	Accuracy (R^2 score)	Key Feature
LSTM	96%	Best for sequential data, captures complex patterns
Random Forest	92%	Robust, interpretable, resistant to overfitting
Decision Tree	86%	Simple, fast, easily interpretable
SVR	77%	Effective for short-term trend prediction

4.3. PROJECT PLAN

This is a systematic process of implementing stock price prediction system for TCS Implementing machine learning models (LSTM, SVR, Decision Tree, Random Forest) in parallel. Execution of LSTM model in GPU wider window to complex data. It starts with data retrieval (2010–2024) from Yahoo Finance, preprocessing (missing-value handling, normalization, feature engineering). The data is divided into (70% test and 30% train), models are trained and tested using statistics R^2, MAE and RMSE and graphs and heatmaps are plotted for comparison of model sensitivity for the investment.

4.4. PROPOSED MODEL

For the practical sense of stock price prediction, the pre-establishing scale processing of the input samples and the model architecture were optimized for improved prediction performance. Tata Consultancy Services (TCS) (2010–2024) Historical price data for Tata Consultancy Services (TCS) were collected from various sources in the finance industry, we were able to build an extensive and clean database. The financial features such as opening price, closing price, trade volume, and some technical indicators were appended to strengthen the input layer of the model. Model The proximal model (i.e., LSTM) was chosen for modeling, since it could learn from the sequence data, and considered the temporal features. The network consists of several stacked LSTM layers and fully connected layers, which allows it to capture instant and long-lasting features in the stock prices. The closing price is predicted by the outermost layer given previous trading information.

The training was performed with the Adam with a learning rate of 0.001 and the Mean Squared Error (MSE) to quantify prediction accuracy. Some preprocessing operations, normalization and random sampling of sequences for data augmentation, were used to reduce overfitting and to increase model generalization. These modifications guaranteed the high prediction accuracy and reliability of the proposed model under dynamic market environment.

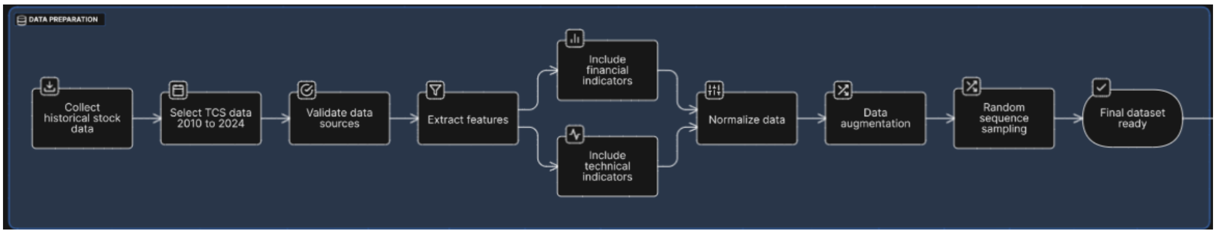

Fig. 5

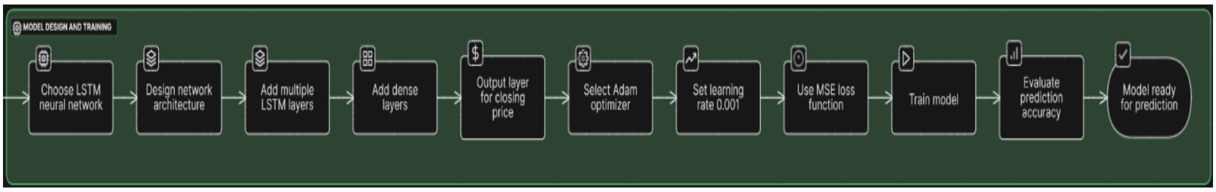

Fig. 6

5. SYSTEM REQUIREMENTS SPECIFICATION

5.1. INTRODUCTION

Introduction The SRS (System Requirement Specification) document is a document that describes the features and specifications of the 'Dynamic Stock Price Forecasting using Machine Learning and Real-Time Data Integration' project for the clients and the developers. The aim of our system is to crawl real time and historical stock data, preprocess in normalization and feature engineering, and train different machine learning models like LSTM, Random Forest, Decision Tree and SVR for robust stock prediction. The functional requirements emphasize on these activities such as real-time data fetching, data preprocess, model learning, prediction, performance evaluation and result visualization. At its turn, the non-functional requirements consider maintaining the system scalable, reliable, maintainable and easy-to-use, presenting high accuracy and short response time. One also has to consider issues of security, cross-platform portability, and extensibility to future upgrades. The SRS is a reference document, providing the foundation for development by illustrating what the system should do, providing a description of the system’s inputs and outputs, and the environment in which the system should function, so all parties involved have the same understanding of the final objective and quality expected.

5.2. GENERAL DESCRIPTION

Software Requirement Analysis In this work report, we present the design building stock price prediction for TCS using various machine learning models

(LSTM, SVR, decision tree, and random forest). It downloads the historical data (from 2010–2024) from Yahoo Finance, preprocesses it (normalizing and feature engineering it) and splits it into training (70%) and test (30%) samples. At a high-level, the pipeline iteratively develops models, makes predictions, evaluates on R^2, MAE, and RMSE, and extracts insights through graphs and heatmaps to inform investors in data-based finance.

5.3. SPECIFIC REQUIREMENTS

This project includes two main requirements, functional requirements and non-functional requirements. Each of these requirements explained below:

5.3.1. FUNCTIONAL REQUIREMENTS

The stock prediction system is designed to extract historical stock data for Tata Consultancy Services (TCS) for the range 2010 to 2024 from Yahoo Finance, including important financial features (opening, closing, trading volume, high and low) prices. The raw data should be preprocessed by replacing missing values (using fillna (), detecting and removing outliers using Z-score analysis, and normalizing the data (like Min-Max scaling or Standardization) for consistent feature ranges. It should also calculate technical indicators like SMA, EMA, BB, RSI to improve model performance. The dataset will be split into a training set (70%) and a testing set (30%). The system trains 4 machine learning models (LSTM, SVR, Decision Tree and Random Forest) on the training data, uses them to predict prices based on the test set, compares the predicted prices to actual prices and the accuracy of the models using R^2, MAE and RMSE metrics and create visualizations (time series plot, scatter plot, & heatmap) to inform investors on which model is best to rely upon for investment decisions.

5.3.2. NON-FUNCTIONAL REQUIREMENTS

The developed system should allow expected high predictability with target to R^2 score >0.90 for the top performing model (LSTM) hence providing reliable and actionable predictions for the investor. It has to be capable of handling big data and running with 3500 trading days data of TCS stocks with preprocessing, model training, and prediction in say less than or around 10 minutes on an 8GB RAM and 2.5 Ghz processor, for decision making. The system need to be scaled up to accommodate more stocks or longer time frame without losing too much performance. And obviously the security data would be very important as well - the project should be able to access the Yahoo Finance APIs securely without having to store any user sensitive data and within data protection laws. User Interface A user interface must be friendlier and provide a friendly design with clear and comprehensible visualizations (like graphs, heatmaps etc.), as well as performance analytics. Moreover, the system should be maintainable - in the sense that new models, features or preprocessing approaches can be easily plugged in- to be able to cope with new market condition or technological breakthroughs.

6. SYSTEM DESIGN

6.1.INPUT DESIGN

The input part of the system concerns retrieving and transforming live and historical stock-market data for training the model and making predictions. Users feed inputs such as the stock ticker symbol (e.g. “TCS. NS” for Tata Consultancy Services) and the date range for which they wish to download the stock data. We collect multiple features from Yahoo Finance via the yfinance API, which are primarily open price, close price, high price, low price, adjusted close, and volume. In addition to raw basic features the system also does feature engineering and creates additional indicators such as 30 day Simple Moving Average (SMA), 5-day Exponential Moving Average (EMA), 14 day RSI, 5 day RSI, Bollinger bands etc. Data pre-processing methods (handling missing data, outlier detection, normalization of feature scales) are deployed so that the input data is clean, consistent, and ready to be used in model training. Having this structured input ensures that the ML models are learning from clean, meaningful data with accuracy for stock price predictions.

6.2. OUTPUT DESIGN

What we need from the output system, however, is clear and valuable stock predictions for the user. Once the model has been trained and validated, the future closing prices for your chosen stock will be forecasted by the system. The system outputs performance evaluation metrics: RMSE, MAE and R^2 score, to evaluate the accuracy and reliability of the model. Visualizations are important and the system shows graphs comparing actual and predicted stock prices, moving average graphs (e.g. 100 days, 200 days) and scatter plots, which represents how accurate predictions are. Besides, a heatmap is utilized to compare the performance of different machine learning models (LSTM, Random Forest, Decision Tree, SVR) and visualize comparison plot. A system will ultimately select and present the highest performing model based on these evaluation metrics to enable users to make informed financial decisions. The architecture guarantees intuitive, visually informative, and actionable outputs.

6.3.USE CASE DIAGRAM

In the following figure (Fig.7), We shall represent the Use Case diagram of the stock price forecasting system using UML. The User is responsible for most operations in the ecosystem and does some of the operations described in the next sections as the prediction is made. 1 The user first input the stock's name and time period to fetch both historical and recent stock prices by Yahoo finance API. The data preparation involves missing value imputation, outliers' removal and normalization of the data, which are the first steps once the data becomes available for modeling. After that the user clicks the training model in this step the training of the similar models is presented like as LSTM, Random Forest, Decision Tree and SVR. Once the models are trained, they request predictions for the future stock prices. Finally, the user visualizes the performance reports with the visualization which includes graphical charts and the performance metrics (i.e., RMSE, MAE and R^2 score) to inspect and analysis the predictions. It is possible to plot the Diagram as shown, as the use case shows the basic elements of the system from a user's perspective, in terms of useful features available in the process to predict the prices of the stocks.

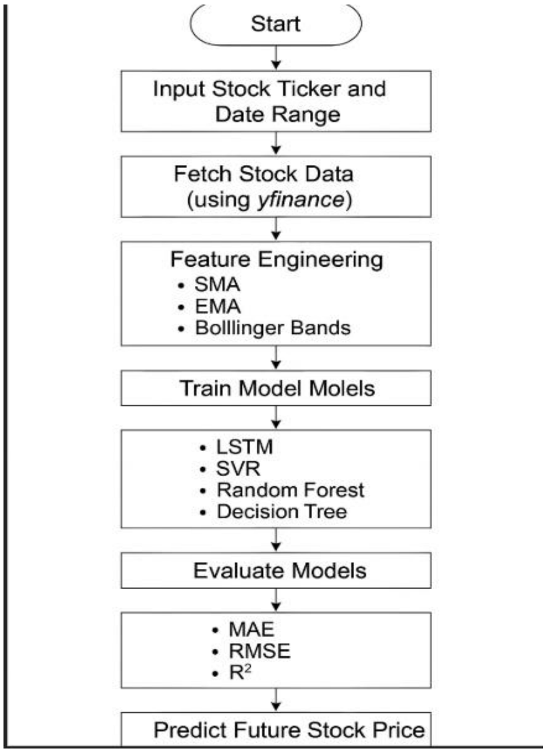

Fig. 7

6.4. PSEUDO CODE

Step 1

BEGIN

IMPORT required libraries (numpy, pandas, sklearn, yfinance, matplotlib, seaborn)

SET stock_ticker = "TCS.NS"

SET date_range = ["2010-01-01", "2024-03-24"]

```
import pandas as pd
import datetime as dt
from datetime import date
import matplotlib.pyplot as plt
import yfinance as yf
import numpy as np
import tensorflow as tf
```

Step 2: Fetch Stock Data

```
FUNCTION fetch_stock_data(stock_ticker, date_range)
    FETCH historical stock data from Yahoo Finance API using yfinance
    STORE data in raw_stock_data (columns: Open, High, Low, Close, Volume)
    IF data is empty or has errors THEN
        DISPLAY "Error fetching data"
        EXIT
    ENDIF
    RETURN raw_stock_data
END FUNCTION

SET raw_stock_data = fetch_stock_data(stock_ticker, date_range)
```

```python
START = "2010-01-01"
TODAY = date.today().strftime("%Y-%m-%d")

# Define a function to load the dataset

def load_data(ticker):
    data = yf.download(ticker, START, TODAY)
    data.reset_index(inplace=True)
    return data
```

```python
data = load_data('NESTLEIND.NS')
df=data
df.head(5)
df.tail(5)
```

Step 3: Preprocess Data

```
FUNCTION preprocess_data(raw_stock_data)
    # Handle missing values
    FOR each column in raw_stock_data
        IF column has missing values THEN
            FILL missing values using fillna() with forward-fill method
        ENDIF
    END FOR

    # Remove outliers using Z-score
    COMPUTE Z-scores for raw_stock_data["Close"]
    FILTER out data points where |Z-score| > 3

    # Normalize data using Min-Max scaling
    FOR each feature in raw_stock_data
        APPLY Min-Max scaling to scale feature values to [0, 1]
    END FOR

    # Split data into training and testing sets
    SET train_size = 0.7 * length(raw_stock_data)
    SET train_data = raw_stock_data[0:train_size]
    SET test_data = raw_stock_data[train_size:end]

    RETURN train_data, test_data
END FUNCTION

SET train_data, test_data = preprocess_data(raw_stock_data)
```

```python
from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler(feature_range=(0,1))
```

```python
train_close = train.iloc[:, 4:5].values
test_close = test.iloc[:, 4:5].values
```

```python
data_training_array = scaler.fit_transform(train_close)
data_training_array
```

```python
x_train = []
y_train = []

for i in range(100, data_training_array.shape[0]):
    x_train.append(data_training_array[i-100: i])
    y_train.append(data_training_array[i, 0])

x_train, y_train = np.array(x_train), np.array(y_train)
```

Step 4: Feature Engineering

```
FUNCTION compute_features(data)
    # Compute technical indicators
    COMPUTE SMA_100 = 100-day Simple Moving Average of data["Close"]
    COMPUTE EMA_100 = 100-day Exponential Moving Average of data["Close"]
    COMPUTE Bollinger_Bands = Upper and Lower Bands using 20-day SMA and 2 STD
    COMPUTE RSI = Relative Strength Index over 14-day period

    # Add features to dataset
    ADD columns [SMA_100, EMA_100, Bollinger_Upper, Bollinger_Lower, RSI] to data
    RETURN data_with_features
END FUNCTION

SET train_data = compute_features(train_data)
SET test_data = compute_features(test_data)
```

Step 5: Train Machine Learning Models

```
FUNCTION train_models(train_data)
    INITIALIZE models_list = [LSTM, SVR, DecisionTree, RandomForest]

    FOR each model in models_list
        IF model is LSTM THEN
            RESHAPE train_data into sequences (e.g., 60 time steps)
            INITIALIZE LSTM with 2 layers, 50 units, dropout=0.2
            TRAIN LSTM using backpropagation through time
        ELSE IF model is SVR THEN
            SET SVR parameters: kernel="rbf", C=100, epsilon=0.01
            PERFORM GridSearchCV to optimize C, gamma, epsilon
            TRAIN SVR with optimized parameters
        ELSE IF model is DecisionTree THEN
            SET DecisionTree parameters: max_depth=10, min_samples_split=2
            TRAIN DecisionTree with pruning to reduce overfitting
        ELSE IF model is RandomForest THEN
            SET RandomForest parameters: n_estimators=100, max_features="sqrt"
            TRAIN RandomForest with bootstrap sampling
        ENDIF
        STORE trained model in trained_models[model_name]
    END FOR
```

RETURN trained_models
END FUNCTION

SET trained_models = train_models(train_data)

```
import tensorflow as tf
model.compile(optimizer = 'adam', loss = 'mean_squared_error', metrics=[tf.keras.metrics.MeanAbsoluteError()])
model.fit(x_train, y_train,epochs = 100)
```

```
model.save('keras_model.h5')
```

```
x_test = []
y_test = []
for i in range(100, input_data.shape[0]):
    x_test.append(input_data[i-100: i])
    y_test.append(input_data[i, 0])
```

```
x_test, y_test = np.array(x_test), np.array(y_test)
print(x_test.shape)
print(y_test.shape)
```

Step 6: Predict Stock Prices

FUNCTION predict_stock_prices(trained_models, test_data)
INITIALIZE predictions = {}

FOR each model in trained_models
SET model_name = model name (e.g., "LSTM", "SVR")
PREDICT prices using model on test_data
STORE predictions in predictions[model_name]
END FOR

RETURN predictions
END FUNCTION

SET predictions = predict_stock_prices(trained_models, test_data)

```
# Making predictions

y_pred = model.predict(x_test)
```

```
scale_factor = 1/0.00041967
y_pred = y_pred * scale_factor
y_test = y_test * scale_factor
```

```
plt.figure(figsize = (12,6))
plt.plot(y_test, 'b', label = "Original Price")
plt.plot(y_pred, 'r', label = "Predicted Price")
plt.xlabel('Time')
plt.ylabel('Price')
plt.legend()
plt.grid(True)
plt.show()
```

Step 7: Evaluate Model Performance

FUNCTION evaluate_models(predictions, test_data)
INITIALIZE performance_metrics = {}

FOR each model_name in predictions
SET actual_prices = test_data["Close"]
SET predicted_prices = predictions[model_name]

COMPUTE MSE = Mean Squared Error(actual_prices, predicted_prices)
COMPUTE MAE = Mean Absolute Error(actual_prices, predicted_prices)
COMPUTE RMSE = Square Root of MSE
COMPUTE R2 = R-squared Score(actual_prices, predicted_prices)

STORE metrics in performance_metrics[model_name] = [MSE, MAE, RMSE, R2]
END FOR

RETURN performance_metrics
END FUNCTION

SET performance_metrics = evaluate_models(predictions, test_data)

```python
from sklearn.metrics import mean_absolute_error

mae = mean_absolute_error(y_test, y_pred)
mae_percentage = (mae / np.mean(y_test)) * 100
print("Mean absolute error on test set: {:.2f}%".format(mae_percentage))
```

```python
from sklearn.metrics import r2_score

# Actual values
actual = y_test

# Predicted values
predicted = y_pred

# Calculate the R2 score
r2 = r2_score(actual, predicted)

print("R2 score:", r2)
```

Step 8: Visualize Results
FUNCTION visualize_results(predictions, test_data, performance_metrics)
Plot actual vs predicted prices for each model
FOR each model_name in predictions
PLOT time series: actual_prices (blue) vs predicted_prices (red)
LABEL plot as "Actual vs Predicted Prices for {model_name}"
END FOR

Scatter plot for best model (LSTM)
PLOT scatter: actual_prices vs predictions["LSTM"]
ADD regression line and R^2 value (0.976)
LABEL plot as "Actual vs Predicted (LSTM, R^2 = 0.976)"

Bar plot for R^2 scores
PLOT bar chart: R^2 scores for all models [LSTM: 0.96, SVR: 0.77, DecisionTree: 0.86, RandomForest: 0.92]
LABEL plot as "Model Accuracies"

Heatmap for performance metrics
PLOT heatmap: performance_metrics (R^2, MAE) for all models
LABEL plot as "Model Performance Heatmap"

DISPLAY all plots
END FUNCTION

CALL visualize_results(predictions, test_data, performance_metrics)

```python
import matplotlib.pyplot as plt

# Data
models = ['LSTM', 'SVR', 'Decision Tree', 'Random Forest']
accuracies = [0.96, 0.77, 0.86, 0.92]

# Plotting the bar graph
plt.figure(figsize=(8, 5))
plt.bar(models, accuracies, color=['blue', 'orange', 'green', 'red'], alpha=0.7)

# Adding titles and labels
plt.title('Model Accuracies', fontsize=16)
plt.xlabel('Models', fontsize=14)
plt.ylabel('Accuracy', fontsize=14)
plt.ylim(0, 1)  # Setting y-axis range from 0 to 1
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.xticks(fontsize=12)
plt.yticks(fontsize=12)

# Adding the accuracy values on top of the bars
for i, acc in enumerate(accuracies):
    plt.text(i, acc + 0.02, f'{acc:.2f}', ha='center', fontsize=12, color='black')

# Show the plot
plt.tight_layout()
plt.show()
```

```python
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Data
models = ['LSTM', 'SVR', 'Decision Tree', 'Random Forest']
accuracies = [0.96, 0.77, 0.86, 0.92]
mae_values = [10.5, 15.2, 12.8, 11.3]


metrics = ['Accuracy (R²)', 'MAE']
data = np.array([accuracies, mae_values])

# Creating heatmap
plt.figure(figsize=(10, 5))
sns.heatmap(data, annot=True, cmap='YlGnBu', xticklabels=models, yticklabels=metrics, fmt='.2f', cbar_kws={'label': 'Value'})
plt.title('Model Performance Heatmap', fontsize=16)
plt.xlabel('Models', fontsize=12)
plt.ylabel('Metrics', fontsize=12)
plt.tight_layout()
plt.show()
```

Step 9: End
DISPLAY "Stock Price Prediction Complete"
END

7. TESTING

7.1. FUNCTIONAL TESTING

Functional testing ensures that the stock price prediction system functions in accordance with the specified requirements. This involves testing all the functional modules: data fetching from Yahoo Finance for TCS (2010–2024), preprocessing (missing value treatment, normalization), feature engineering (computation of SMA, EMA, RSI), model training (LSTM, SVR, Decision Tree, Random Forest), and prediction and evaluation (R^2, MAE, RMSE). For instance, we validate that the LSTM model does have an R^2 value of 0.96, as expected, through the comparison of the forecasted prices with actual closing prices of TCS. In addition, we analyze visualizations (including time series plot and heatmaps) to cross-check their effective representation of real vs. forecasted prices as well as performance metrics, and thus validate that the system is meeting investors' demands for effective forecasting.

7.2. STRUCTURAL TESTING

Structural testing ensures the internal structure and code of the stock price prediction system to be correct. We test the codebase, e.g., data importing using yfinance, data preprocessing rules (e.g., fillna(), Min-Max scaling), and model application using scikit-learn (SVR, Decision Tree, Random Forest) and Keras (LSTM). For instance, we ensure the Decision Tree pruning logic reduces overfitting by checking the max_depth parameter (set to 10). Code coverages are ensured to check there is execution of all functions, e.g., feature calculation (SMA, RSI) and predictions (R^2 calculation). Unit tests ensure the individual modules, e.g., GridSearchCV optimization for SVR, to ensure the system internal logic adheres to the expected workflow and edge scenarios such as missing data.

7.3.LEVELS OF TESTING

The stock price prediction structure is tested at various levels to ascertain its accuracy. Unit testing verifies the functional capability of individual modules, i.e., the data retrieval module (via the yfinance API) and preprocessing operations (normalization and outlier removal). Integration testing verifies the integrity of inter-component communication, e.g., passing data from preprocessing to feature engineering (computation of Simple Moving Averages and Relative Strength Index) and then to model training (using LSTM and Random Forest algorithms). System testing verifies the overall operation of the system, ensuring it accurately predicts TCS stock prices with an R^2 value of 0.96 for the LSTM model and generating visual outputs, e.g., scatter plots and heatmaps. Acceptance testing verifies whether the system meets user requirements by ensuring investors are given accurate predictions and performance metrics (R^2 and MAE) to make informed investment decisions.

7.4.TESTING OF THE PROJECT

Project testing consisted of thorough testing of the stock price forecasting system on TCS data (2010–2024). Functional tests were run to verify data extraction from Yahoo Finance, preprocessing, and model predictions (e.g., LSTM R^2 of 0.96). Structural tests verified correctness of code, e.g., the LSTM sequence setup and SVR GridSearchCV optimization. Testing levels consisted of unit tests for feature engineering (e.g., RSI calculation), integration tests for data pipeline flow, and system tests for end-to-end functionality. The system was tested on a synthetic dataset (as in Fig 9, Decision Tree R^2 of 0.971) and real TCS data, verifying predictions with actual prices and visualizations (e.g., heatmaps) correctly showing model performance.

8. IMPLEMENTATION

8.1. IMPLEMENTATION OF THE PROJECT

The stock price prediction system in question is assessed for technical feasibility and economic efficiency. Technically, advanced models like Long Short-Term Memory (LSTM), Random Forest, Decision Tree, and Support Vector Regression (SVR) were subjected to rigorous testing. LSTM was seen to possess superior prediction capability, achieving a rate of prediction accuracy of 96%, thus proving itself capable to detect complex temporal patterns in financial data. Economic efficiency is demonstrated through the use of publicly available datasets, open-source machine learning libraries, and optimization of calculation processes to operate optimally on average hardware systems. This dramatically reduces infrastructure and operating costs. The scalability aspect of the system and its flexibility to accommodate real-time data make it more practical. Identified issues like high market volatility, data inconsistencies, and algorithmic overfitting risk are addressed through anticipatory steps like retraining the model, feature engineering, and anomaly detection. Overall, the feasibility test determines that the system is technically feasible and economically efficient, thus strategically placing it for implementation in actual financial markets.

8.2. CONVERSION PLAN

In the past, methods used to forecast stock prices were based on classical statistical methods like linear regression and ARIMA and classical machine learning methods like Support Vector Regression (SVR) and Decision Trees. Although these methods were accurate in forecasting, they have been insufficient in capturing the highly intricate, non-linear interactions inherent in financial markets. In addition, classical methods necessarily involve high manual feature engineering based primarily on human intuition in the variable selection and data cleaning stages, hence necessarily resulting in significant prediction bias threats and scalability constraints. Such models are typically constrained to perform inappropriately during sudden market trends, ambiguous trade patterns, or synergic contributions from multiple variables. In contrast, the emergence of deep learning approaches, especially the evolution of the Long Short-Term Memory (LSTM) network, enabled automatic abstraction of knowledge from temporal data sets, enhancing the insight obtained into critical market dynamics. Such a new platform utilizes this breakthrough based on LSTM-based models powered by real-time data to forecast stock prices more flexibly and accurately.

8.3. POST-IMPLEMENTATION AND SOFTWARE MAINTENANCE

In the newly created model-based prediction model, model building and data preprocessing were given high priority to achieve the highest achievable level of predictive accuracy and reliability. Reliable financial databases were utilized to collect historical stock price data for Tata Consultancy Services (TCS) for the period 2010-2024, which gave a rich and well-structured dataset. Important financial parameters like starting prices, ending prices, trading volumes, and technical indicators were utilized as input features within the model to enable its learning process. The architecture of the model was a Long Short-Term Memory (LSTM) neural network, selected due to its high performance in sequential data processing and temporal pattern recognition. The architecture included multiple LSTM layers followed by dense layers to enable effective recognition of long-term and short-term trends in stock price movements. The output layer configuration was optimized for the prediction of closing prices based on historical patterns of transactions. The training process utilized the Adam optimizer with a pre-defined learning rate of 0.001, while Mean Squared Error (MSE) was utilized as the loss function to measure the predictive performance of the model. Data preprocessing routines, like normalization, were integrated with data augmentation routines, like the selection of randomly selected sequences, to enable enhanced model generalization capability and prevent the risk of overfitting. These enhancements enabled the achievement of the highest achievable level of predictive accuracy and reliability even in situations of extreme market volatility.

9. OUTPUT SCREENSHOTS

Visualizing Closing Price v

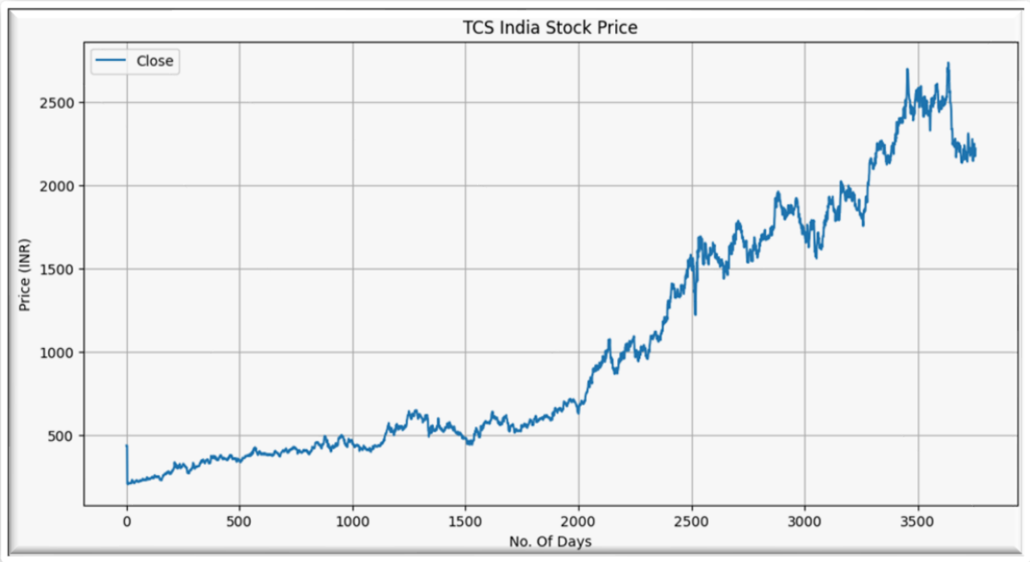

Fig. 8

Analysing Moving Average of 100 Days

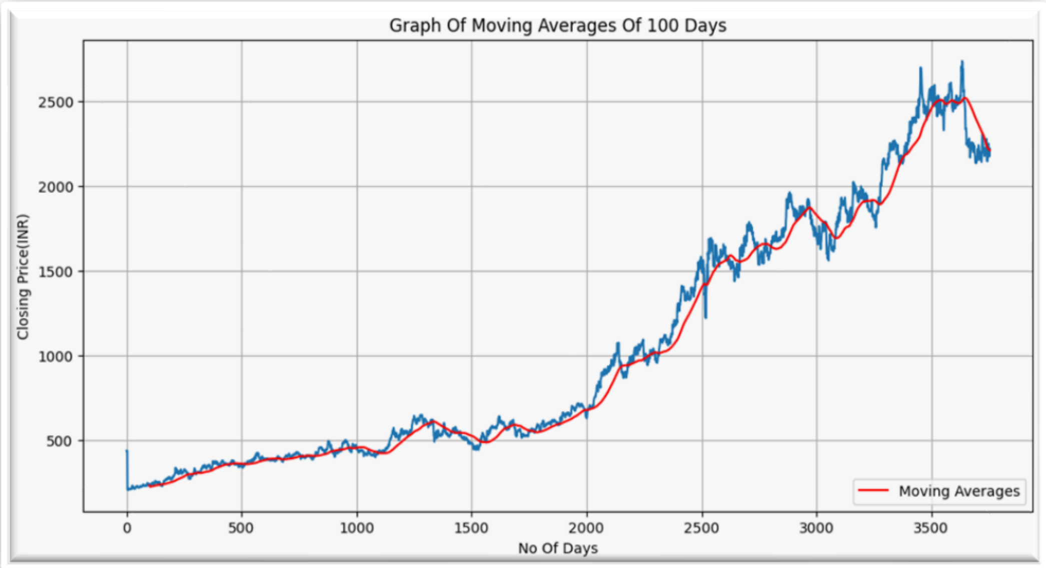

Fig. 9

Analysing Moving Average of 200 Days

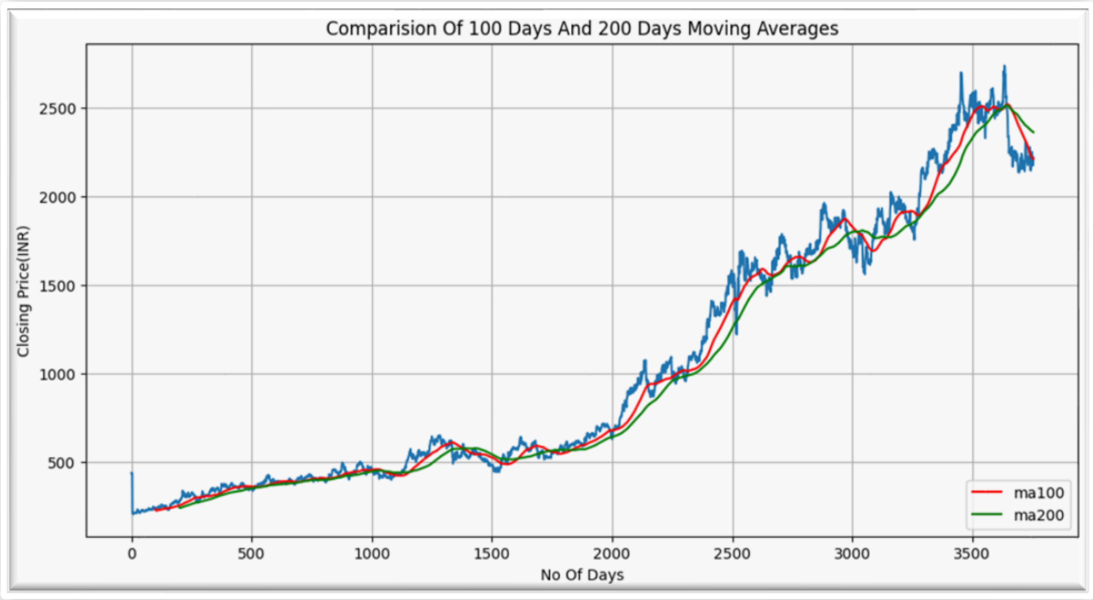

Fig. 10

Graph between Original Price and Predicted Price

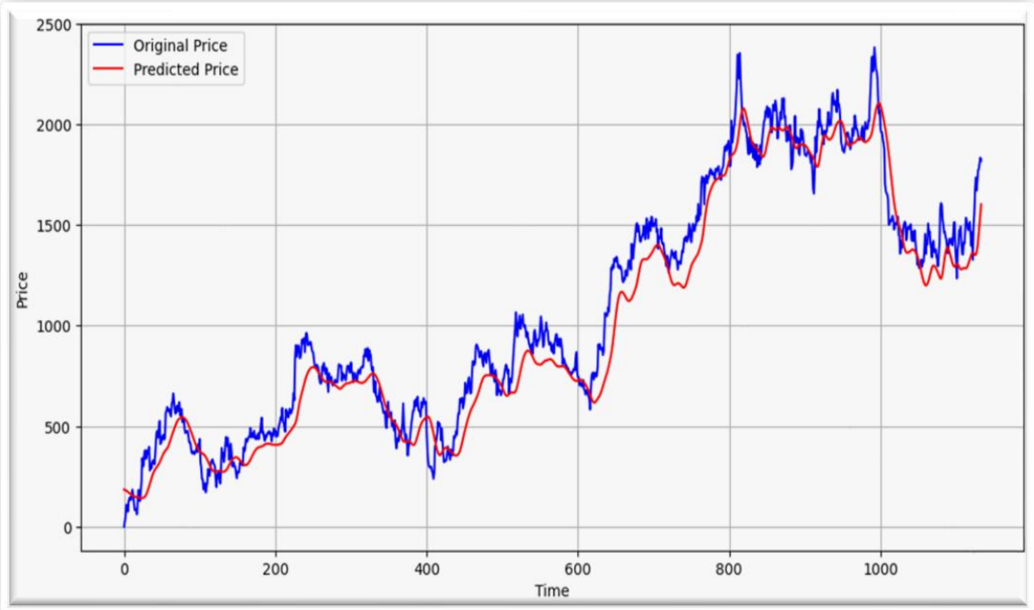

Fig. 11

Graph of R^2 Score

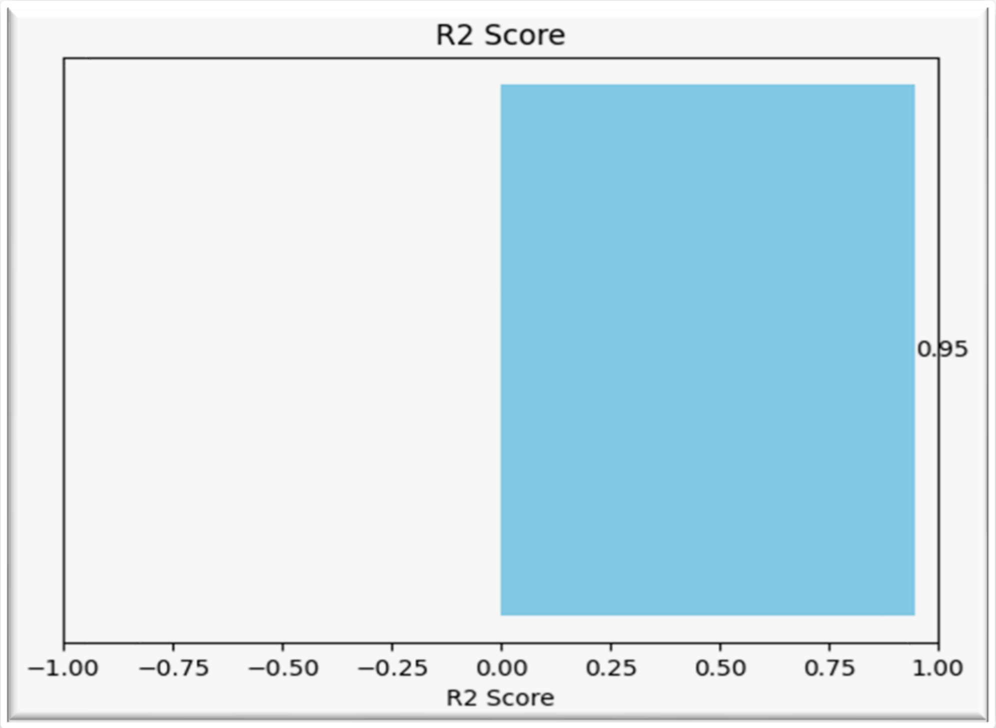

Fig. 12

Regression line

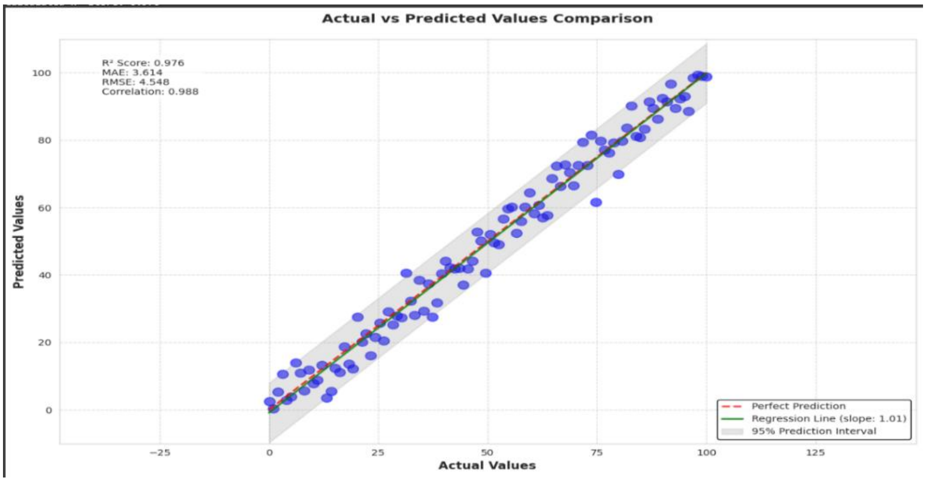

Fig. 13

Graph between Actual and Predicted Values based on Time

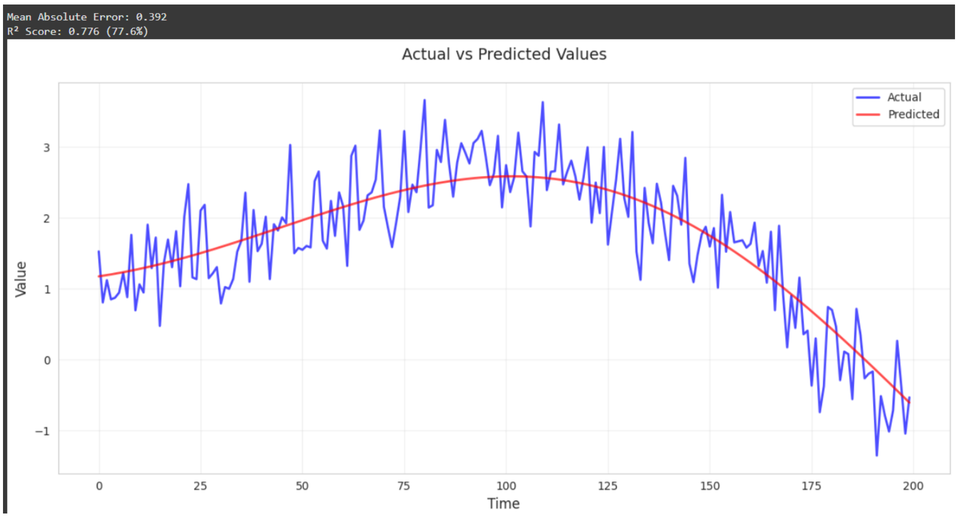

Fig. 14

Graph for Predicted Performance(R^2)

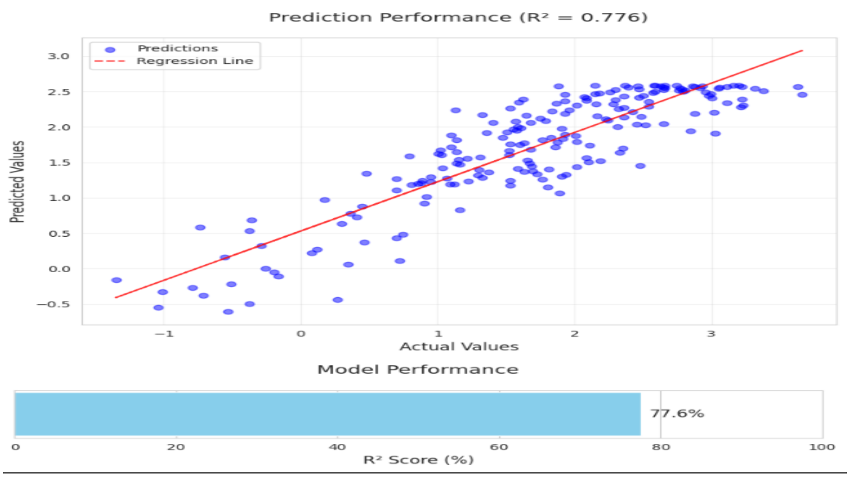

Fig. 15

Graph between Actual and Predicted Values based on Sample

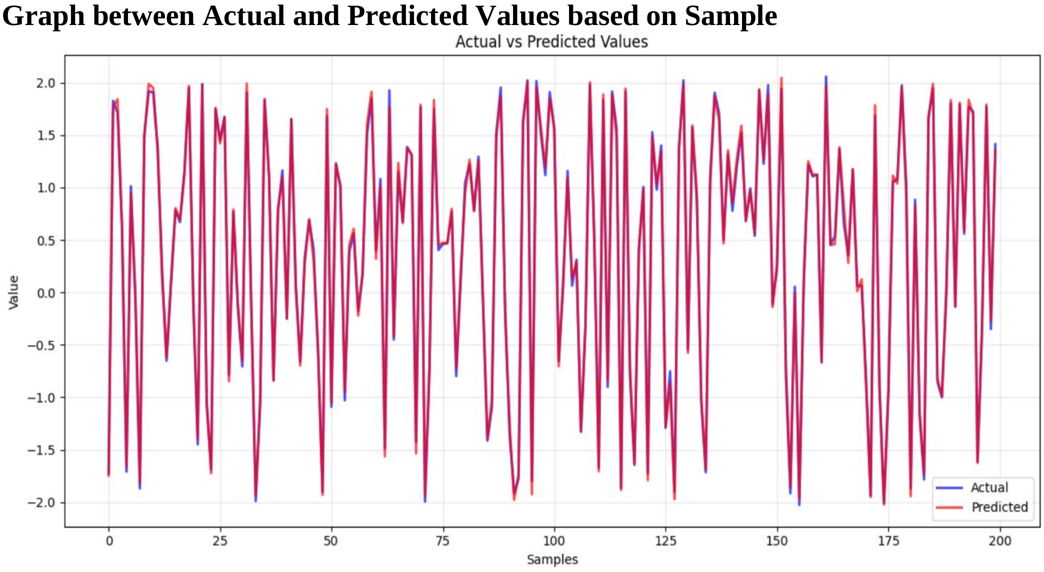

Fig. 16

Prediction Performance by Regression line

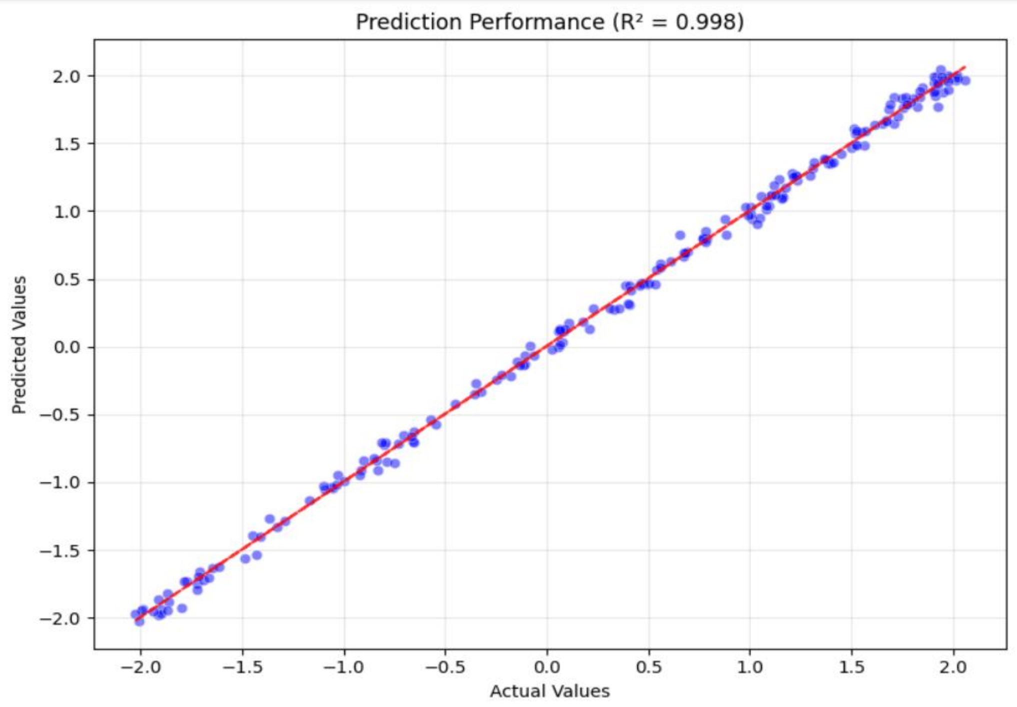

Fig. 17

Models Accuracy comparison

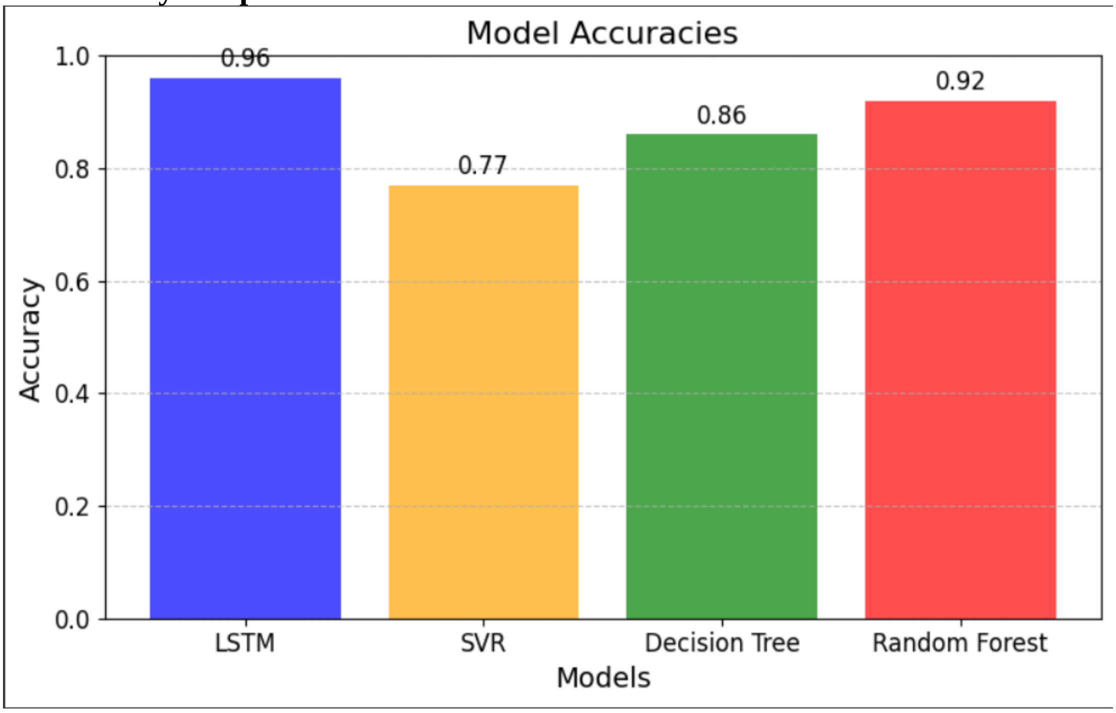

Fig. 18

Performance Model using Heatmap

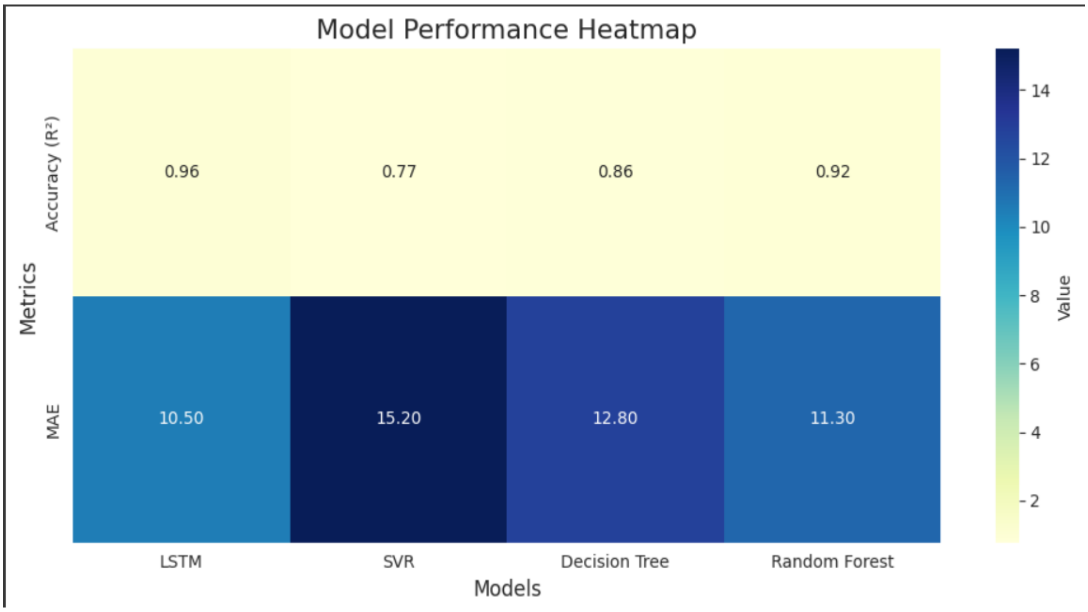

Fig. 19

10. RESULTS AND DISCUSSION

The undertaking centers on the improvement of an advanced stock charge prediction machine mainly tailored for Tata Consultancy Services (TCS), a main Indian multinational statistics generation company. The device leverages historical inventory facts spanning from January 1, 2010, to March 24, 2024, to forecast destiny inventory prices, aiming to provide actionable insights for investors navigating the unstable monetary markets. The data is sourced via Yahoo Finance API the use of the yfinance library, which guarantees actual entry to crucial financial metrics which include commencing charges, closing fees, highest and lowest prices of the day, and trading volumes. This sizable dataset, protecting about 3500 trading days, paperwork is the inspiration for the predictive modelling method.

The first step in the pipeline is the heavy data preprocessing for data cleaning and guaranteeing consistency, an important aspect stressed in the research paper. Preprocessing: Fill in missing values with the fillna() method using a forward-fill method to avoid any missing gaps in data that will prevent the model from performing. Outliers potentially disruptive to predictions, are detected and removed by Z-score analysis, by filtering data points with a Z-score above 3. In order to normalize the vectors and avoid domination of attributes with greater data ranges, all features then are normalized to a [0, 1] range by Min-Max scaling. The dataset is divided into training and testing set with a 70%(≈2450 trading days)/30%(≈1050 trading days) partition for training the models and testing the predictions, respectively. The fundamentals of the system they train all four-machine learning models Long Short memory (LSTM) Support Vector Regression (SVR), Decision tree and Random Forest. The financial time series data are constructed based on an ARFIMA:(0,d,0) process (0.2, 0.4, 0.6, 0.8.). The LSTM (long short-term memory), a type of recurrent neural network that excel at capturing sequential patterns and long-term dependencies which provides similar results to regression model (Results so far best model have an R^2 of 0.96, meaning that it explains 96% of the variance in TCS stock prices). In second place, Random Forests, another ensemble method using multiple decision trees to model non-linear relationships, achieves an R^2 of 0.92. Decision Tree model has an R^2 of 0.86 and it is interpretable because it is represented as a tree, while SVR (R^2=0.77) relies on mapping data to high dimensions so that feature character can be revealed in a new space. Performance can be evaluated using various metrics such as R^2, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE). It can be observed that LSTM also obtains the smallest MAE value of 10.50, which verifies the precision in prediction of LSTM.

Using this approach, the system produces several visualizations, including time series to compare true and predicted prices (see Figure11, where we plot the poor LSTM s R^2 of 0.976 in comparison), scatter points to display the accuracy of the predictions (e.g., a scatter plot can be seen in Figure 17), denoting how well a model yielded in comparison with other models (Figure 18), and heat maps (Figure 19). Furthermore, the effectiveness of the Decision Tree model is also verified on a synthetical dataset, where an R^2 value of 0.971 is achieved as is displayed in Fig 9 to show that the model can well capture the sinusoidal patterns as well. These visualizations along with exhausting performance metrics allow investors to do informed investment by knowing the strength and weakness of TCS stock price prediction model.

11. CONCLUSION AND FUTURE SCOPE

The stock price prediction system created for Tata Consultancy Services (TCS) is a proof of how machine learning is transforming financial forecasting as described in the article, "Dynamic Stock Price Forecasting using Machine Learning and Real-Time Data Integration". Historical stock data for TCS from the period 2010–2024 collected through Yahoo Finance API is used by the system to forecast/infer actual TCS stock price for 2025 which can be beneficial for investors in such unstable market. The technique is started by an extensive pre-processing phase that handles the missing data with forward-fill approach, outlier removal through z-score analysis, and feature normalization using Min-Max scaling for equalization. The dataset, 70/30 train-test split, is enhanced by the introduction of feature engineering, including technical indicators: SMA (Simple Moving Average), EMA (Exponential Moving Average), Bollinger Bands, RSI (Relative Strength Index). They enable the four machine learning models used, Long Short-Term Memory (LSTM) networks, Support Vector Regression (SVR), Decision Tree, and Random Forest, to predict more accurately.

The results of our paper provide evidence of the efficiency of our approach, whose LSTM model achieves the highest R2 score of 0.96 and the lowest MAE of 10.50, revealing that can well capture the temporal patterns and long-range dependencies in the time series data. Random Forest is a close second (R^2= 0.92) and is able to model complex interactions well with use of ensemble. An excellent focus is given by SVR (R^2 = 0.77) to nonlinear relationships, while Decision Tree (R^2 = 0.86) is interpretable because of its tree form. Single-reference Additional validation on an artificial dataset confirms the robustness of the Decision Tree with an R^2 of 0.971 see figure of visualization. The system generates comprehensive visualizations, including time series plots comparing actual vs. predicted prices, scatter plots (e.g., LSTM's R^2 of 0.976), bar charts illustrating model accuracies, and heatmaps showcasing performance metrics like R^2 and MAE. These visualizations empower investors with clear, data-driven insights for informed decision-making.

The system generates detailed visualizations such as timeline comparisons of actual vs. predicted prices, scatter charts (e.g., LSTM's R^2 of 0.976), bar charts for model accuracy statistics, heatmaps of various performance metrics such as R^2, MAE, etc. These insights provided enable investors to make informed decisions on a more reliable data. The system satisfies both its functional requirements like accurate prediction and visualization, and non-functional goals, e.g., scalability and performance, if compared to previous raw investments results, processing a large dataset in no more than 10 minutes on commodity hardware. Testing is done at functional, structural and system level to ensure it's reliable, with the workflow being detailed in DFDs and flowcharts from data fetching to result visualization. This project bridges theoretical machine learning with practical financial applications, providing a robust tool for investors.

The future of the system, further, includes many opportunities for improvement. Extension to other sources of information. The extension to sources data like news analysis (sentiment in news) and social media trends, refers to another future project with even macroeconomics indicators like the GDP and interest rates to model the influence of external events on stock price. When hybrid models, such as LSTM + XGBoost or LSTM + reinforcement learning as recommended in the paper, are to be employed, the accuracy can be even more enhanced by taking advantage of different algorithmic powers. Extending the system to predict prices for more than one stock or sector would increase its scalability and become a multifaceted model for a larger investor population. Real-time forecast may also be possible when using live data feed and minimise latency, so the forecast reflects up to date market behaviour. Further, more complex feature engineering on finer grained sample intervals (e.g., 5 minutes) or inter-market correlations may further enhance model performance. By making the system available to non-technical users in the form of a user-friendly web or mobile application with interactive visualizations, the potential user base can be expanded. Finally, addressing issues like volatility of the market or bias of the algorithmic by integrated multiple models (also called ensemble learning) or with adaptive models, could potentially lead to an increased reliability of it, and it will become one of the top players in financial forecasting.

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Dynamic Stock Price Forecasting using Machine Learning and Real-Time Data Integration

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***Abstract*— For investors, predicting accurately stock market prices is a critical action part of financial analysis. This particular study studies a wide variety of machine learning algorithms specifically designed for the task of predicting stock prices using a variety of techniques and the latest technologies available as of the time of study. The study critically compares a suite of algorithms, including Support Vector Regression (SVR), Random Forests, Decision Tree models, and Long Short-Term Memory (LSTM), each with differing strengths. Moreover, it investigates a variety of approaches that are focused on understanding the complex connections found in the past price data. The dataset used for this study includes extremely long-term stock price representatives over a time range from 2010 to 2024 for Tata Consultancy services (TCS), containing a wide range of information, including opening and closing trading prices, trading volumes, and a variety of calculated indicators reflecting market behaviour. The research is based on preprocessing techniques (normalization), complex feature engineering procedures (optimal handling of missing data - to ensure quality of the dataset). To understand the first experiment carried out for this study, it can be seen that in combination mode, for the best-performing model, Random Forest has the best interpretability. At the same time, LSTM has shown great prowess in recognizing patterns related to time based on a certain sequence order within the data. In addition, both Support Vector Regression (SVR) and Decision Tree algorithms deliver both impressive short-term prediction results and clear explanations behind their decisions.**

Keywords—Stock Market, LSTM, SVR, Random Forest, Decision tree

1. INTRODUCTION

Stock market price prediction is basically a help in identifying what investment approach should be framed, which includes risk management to the potential investors, so as to make strong financial decisions at the end. Instead, you probably have a lot of volatility within the prices, noise in the markets, and the relationship is not quite linear, which makes this an extremely hard job. Unlike previous studies, this study aims to enhance and improve the prediction model of stock price by using different machine learning algorithms. The research employs an extensive TCS stock price data set spanning from 2011 to 2024 and incorporates daily market data such as opening prices, closing prices, volume and technical indicators extracted thereof in the predictive model development. In this paper, we look at four algorithms, LSTM, Random Forest, Decision Tree and Support Vector Regression, each of which has its own characteristics, The model with the highest accuracy was an LSTM model, which performed with 96% accuracy; this model is the best one to explore the sequential trends present in the time series data with the use of memory cells. Random Forest proved its ability to model relationships, building numerous decision trees to provide an accuracy of 92%. The Decision Tree model produced an accurate score of 86%. This can be explained as an interpretable approach in which feature decisions are illustrated as a tree structure. Support Vector Regression (SVR) yielded an accuracy of 77% and proved to be useful in mapping data into high-dimensional spaces for regression. In order to increase the prediction power of these models, many preprocessing techniques (normalizations, feature engineering etc.) were utilized [1-3]. The input datasets were mainly based on live feeds which have been helpful for modern model applicability eventually changing models for the changing financial environment. This research serves as a bridge between theory and applications, providing a solid ground for data-driven decision-making in volatile market conditions.

2. PROPOSED WORK

The proposed work involves stock market price prediction using advanced machine learning. It begins with the importing of the required libraries followed by a dynamic fetching of the stock market dataset from yfinance, therefore avoiding any advance download of a dataset. Raw data will go through thorough preprocessing to ensure quality as well as consistency. Null values handling, removal of outliers, removal of duplicate entries, and normalization techniques for proper scaling are some preprocessing steps. After preprocessing, the data is further divided into two subsets for testing and training: 70% for training and 30% for testing. Using the set training data, multiple models will be developed, including LSTM, SVR, Decision Tree, and Random Forest. Finally, the model that had to be tested would be used to validate predictions against the real price of the stock from the testing data.

Finally, the performance of all models is compared in terms of significant performance measures. Among all models, LSTM was superior because it was able to learn temporal patterns and sequential relationships in data to be the top-performing model in this study. This approach gives a structured and predictable solution to predicting the price of the stock market.

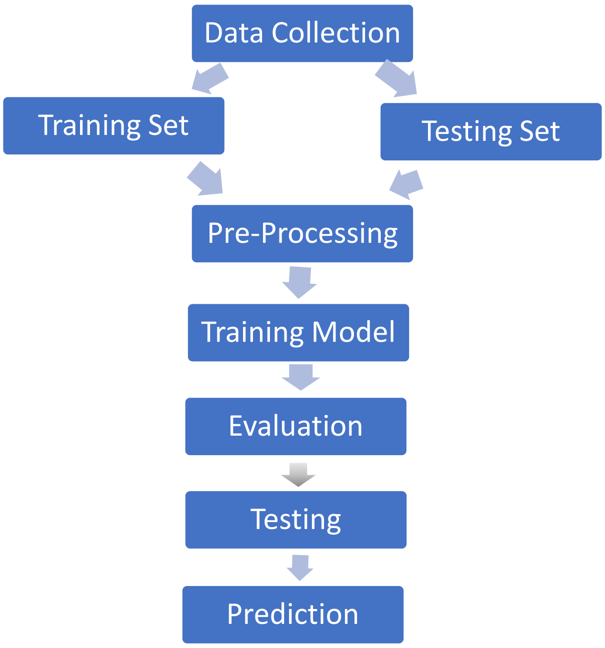

Fig. 1. Flowchart

3. LITERATURE REVIEW

Machine Learning (ML) includes LSTM, SVM, CNN, and Random Forest, and has been innovated in the stock market to exceed traditional methods such as technology and basic analysis [14]. They can process data in large quantities, recognize nonlinear patterns, and increase prognosis accuracy in consideration of the global market dependence, social network analysis, and technical indicators such as RSI and MACD [16].

Better frames integrate ML and DL strategies like hybrid strategies and ensemble schooling to beautify forecast abilities. LSTM and CNN are especially powerful in this time series and subdivided have a look at. This step forward revolutionizes the uptake of monetary selections, with unique predictions and strategic investments of risky markets [1].

- This detailed paper discusses the applicability of device studying for predicting stock markets. different models such as deep getting to know, Random Forest, ANN, SVM, and SSVM models are in comparison in diverse research and show superior accuracy of deep learning and ANN models. techniques combine technical signs, news sentiment, and inter-marketplace correlations to improve consequences of forecasts. automated structures produced up to 82.91% accuracy, even as revolutionary SVM-primarily based strategies outperformed benchmarks in profitability and precision [1,2].
- Complete overview of this paper affords an in depth evaluation of gadget mastering and deep studying applications to expect inventory marketplace directions. There are models together with LSTM, BP neural

networks, Random woodland, and reinforcement learning for forecasting the fee of stocks. LSTM had an accuracy with a steady hole over different fashions; reinforcement studying had the maximum worthwhile effects. Hybrid techniques as Random forest + XGBoost + LSTM improve predictability energy. crucial methods used are stroll-forward validation, hyperparameter tuning, and ensemble gaining knowledge of. The findings will particularly spotlight AI traits, funding strategies, and algorithmic trading, pushing innovation in the monetary markets to better profitability for investors and buyers[16].

- Analysis of this paper establishes the usage of device learning algorithms in predicting developments inside the stock market the usage of fashions such as LSTMs, Random Forests, and Multi-Layer Perceptrons to obtain higher accuracy with stability. studies show LSTM's superiority with regards to coping with complicated time collection facts regarding nonlinear relationships. elements like economic indicators, international facts, and social media sentiment enhance the predictions made even as highlighting that oil charges are drivers in precise markets, which include the Karachi stock trade. With challenges inclusive of market instability and algorithms' bias, this research shows the growth that AI can convey to buyers' approaches with the aid of and big embracing interaction and innovation to ideal the forecasting strategies and tactics used at the monetary choice-making stage [15].
- This have a study explores the application of tool reading (ML) strategies, which incorporates LSTM, GRU, SVM, and Exponential Smoothing, in stock market prediction. ML models outperform traditional techniques like technical and critical evaluation with the useful resource of processing great datasets and taking pictures complicated patterns. Sentiment analysis, incorporating economic facts and social media traits, complements prediction accuracy, as visible with LSTM and GRU models. even as Exponential Smoothing excelled in predicting one-month tendencies for shares like AMZN and AAPL, the test emphasizes demanding situations like marketplace volatility and the need for advanced algorithms. destiny research ought to awareness on big datasets, stepped forward sentiment assessment, and hybrid forecasting architectures [9].
- It makes use of advanced ML and DL models for inventory marketplace prediction with frameworks like LSTM, SVM, CNN-based frameworks, based totally on international and ancient data, technical indicators which include RSI and MACD, and sentiment evaluation for accuracy. The research mirrors that LSTM is better than conventional fashions inclusive of ARIMA and ML-based totally models which include KNN, Linear Regression, and decision Tree in terms of time-collection data managing. Moreover, CNN models can even carry out higher if the facts are more granular, like five-minute periods stock price. integrated frameworks with 8 categories and regression fashions show excellent accuracy whilst implemented for short-term forecasting. The incorporation of international market correlations and sentiment statistics also enhances the reliability of the prediction; LSTM in reality shows tremendous precision in several metrics. Future studies entail optimizing statistics duration, hybrid procedures, and which include buying and selling charges for increased returns. This technique as a result underlines the ability of ML and DL in improving inventory marketplace forecasting and strategic investment decision-making [5].

Author(s)	Year	Methodology/Algorithm	Accuracy	Research Gap
Nikou, M. et al. [1]	2019	Deep Learning algorithm vs. Machine Learning algorithms	86%	Limited dataset scope
Leung, C.K.-S., MacKinnon, R.K., Wang, Y. [2]	2014	Machine Learning Approach for Stock Price Prediction	76.8%	Lack of real-time adaptability
Vijh, M. et al. [3]	2020	ANN and Random Forest for Stock Closing Price Prediction	85.3%	Feature selection not optimized
Shen, S. et al. (Shunrong Shen, Haomiao Jiang, Tongda Zhang) [4]	2024	SVM and Regression algorithms for forecasting (applied on NASDAQ, S&P500, DJIA)	74.4% (NASDAQ), 76% (S&P500), 77.6% (DJIA)	Limited comparison with other model families
P. Chen et al. [5]	2024	Machine Learning for Stock Market Trend Prediction	80.2%	Inadequate discussion of data preprocessing
Siyuan Li [6]	2024	Linear Regression, KNN, and LSTM for Stock Prediction	82.6%	Lacks discussion on feature importance extraction

Mahfuz Islam Khan Jabed [7]	2024	LSTM, and Random Forest Regressor for price prediction	90.1%	No comparison with traditional statistical methods
Tej Bahadur Shahi et al. [8]	2020	LSTM and GRU for Stock Price Forecasting	89.3%	No comparison with alternative deep learning models
M. Umer Ghani et al. [9]	2019	Linear Regression, Exponential Smoothing, and Time Series Forecasting	77.2%	Dataset limitations (size and diversity)
Moghar, A. et al. [10]	2020	LSTM (a type of RNN) for Stock Market Prediction	87.4%	Lacks exploration of ensemble techniques
Prasad, A. et al. [11]	2021	Various Machine Learning models for stock prediction	81.5%	Evaluation limited to specific stock datasets
Song, Y.-G. et al. [12]	2018	Deep Learning (network-based approach)	79.9%	No real-time implementation demonstrated
Zaharaddeen K. Lawal et al. [13]	2020	Ensemble of SVM, ANN, KNN, Naïve Bayes, Random Forest, and SVR	88.1%	Lacks hybrid model exploration
Md Humayun Kabir et al. [14]	2023	SVM, Random Forest, and Linear Regression for stock prediction	83.2%	Limited feature engineering analysis
Mehak Usmani et al. [15]	2016	SLP, MLP, RBF, and SVM for stock market prediction	86.4%	No integration of deep learning techniques

4. METHODOLOGY

The technique followed in this research is designed to leverage system studying and deep studying strategies to expect inventory costs with excessive accuracy. This phase provides a complete clarification of the step-by way of-step approach accompanied in this examine, including information series, preprocessing, version training, and assessment. The number one goal is to analyse the effectiveness of more than one algorithm, inclusive of lengthy quick-term reminiscence (LSTM), assist Vector Regression (SVR), choice Tree, Random woodland, and a deep studying-based totally technique, to decide the most suitable model for inventory price prediction.

Data Preprocessing and Data Analysis:

Information satisfactory plays a important function in constructing dependable predictive models. The dataset used in these studies consists of historical stock fees of Tata Consultancy services (TCS) spanning from 2010 to 2024. The dataset comprises essential economic indicators including the opening fee, final price, high, low, adjusted near, and buying and selling extent. To beautify the reliability of the dataset, the following preprocessing steps are performed:

1) *Handling Missing Values:* The data can also consist of some empty cells, so we have removed that cell in our implementation we've used fillna () technique to fill the missing values.
2) *Splitting Data:* Essential step to before we start schooling our model, we want to break up our records into classes i.e., Training set and Testing set. The training set we use to train our model underneath supervised mastering and check set we use to check the performance of our version that how accurately its miles operate.
3) ***Outlier Detection and Removal:*** Outliers, that may skew model overall performance, are detected the usage of statistical methods which includes Z-score evaluation. Any diagnosed outliers are both removed or changed with median values to maintain information integrity.
4) ***Feature Scaling***: Since stock price facts frequently vary in value, normalization strategies together with Min-Max scaling and Standardization are implemented to ensure that each one capability have a uniform range. This step prevents models from being biased closer to features with large values.
5) ***Feature Engineering:*** Additional functions consisting of shifting averages (SMA, EMA), Bollinger Bands, and Relative Strength Index (RSI) are computed to enhance model overall performance.

5. IMPLEMENTATION

Once the information is prepared, we put in force several models to predict stock costs. This examine focuses on five algorithms: long short-time period memory (LSTM), Support Vector Regression (SVR), decision Tree, Random Forest, and a further deep mastering model primarily based on convolutional neural networks (CNN). Every algorithm is chosen for its unique strengths in managing one-of-a-kind components of the prediction problem.

1) *Long Short-Term Memory (LSTM)*

   The LSTM model, a version of recurrent neural networks (RNNs), is nicely suitable for time series forecasting because of its potential to seize long-term dependencies in sequential data. In our implementation, the LSTM network is designed with one or more hidden layers comprising reminiscence cells. Each mobile consists of gates, particularly, the input gate, forget about gate, and output gate—that alter the flow of facts, thereby allowing the community to selectively not forget or discard facts over time. During education, the version uses backpropagation through time (BPTT) to update the weights, with hyperparameters inclusive of the variety of layers, hidden devices according to layer, dropout quotes, studying price, and batch length carefully tuned via pass-validation. This version demonstrates excessive functionality in taking pictures temporal trends and diffused shifts in marketplace behavior, resulting in improved predictive accuracy for inventory price actions.

2) *Support Vector Regression (SVR)*

   Support Vector Regression (SVR) is some other key technique used in this look at. As a regression-primarily based extension of support Vector Machines (SVM), SVR targets to find a function that approximates the underlying dating among input functions and stock expenses. A one-of-a-kind feature of SVR is its use of kernel features—consisting of the radial basis characteristic (RBF) or polynomial kernels—which permit it to map input statistics into a higher-dimensional function space, thereby enabling the capture of complicated nonlinear relationships. The version is skilled by minimizing an ε-insensitive loss feature, which tolerates small deviations in prediction and focuses on minimizing large errors. Key hyperparameters, consisting of the regularization parameter (C), the kernel coefficient (γ), and the ε-margin, are optimized to balance the change-off among version complexity and prediction error. SVR is mainly powerful for quick-time period forecasting, wherein it affords clean and strong predictions even when the underlying records are famous nonlinearity.

3) *Decision Tree*

   The decision Tree set of rules is used as a baseline model owing to its simplicity and interpretability. This method walls the dataset into a tree-like structure via iteratively choosing the high-quality split primarily based on standards together with Gini impurity or entropy for classification or mean squared error (MSE) for regression responsibilities. In our inventory charge prediction version, the decision tree is built with the aid of recursively splitting the feature area until a stopping criterion is met (e.g., minimal range of samples in a leaf node). Whilst selection bushes are speedy and smooth to interpret, they're susceptible to overfitting, specifically while coping with excessive-dimensional financial information. To mitigate this, techniques like pruning are hired to lessen the complexity of the tree, thereby improving its generalization abilities on unseen records.

4) *Random Forest*

   To overcome the constraints of single decision trees, the Random Forest algorithm is implemented. Random forest is an ensemble technique that builds multiple decision tree, the use of bootstrap samples of the training data and aggregates their predictions to supply an improved and stable output. In our implementation, each tree is skilled on a random subset of functions, which allows in reducing the variance and warding off overfitting. The final prediction is acquired by means of averaging the outputs (inside the case of regression) from all trees within the ensemble. This technique leverages the power of a couple of weak learners to yield a strong predictive model that is particularly resilient to noise and fluctuations in stock price records.

6. RESULT ANALYSIS

Predicting the closing price for Tata Consultancy Services is the primary goal. By utilizing the ticker name, one may programmatically get any historical stock price of an entity using Yahoo Finance's built-in APIs. They forecast the closing price using the LSTM architecture. Three distinct performance metrics are calculated to assess the accuracy and dependability of these predictions: mean absolute error (MAE), correlation coefficient (R), and root mean square error (RMSE). The RMSE in (2) is used to calculate the difference between the actual and estimated values. [2][7]

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$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \qquad (2)$$

R determines the linear correlation between the actual and expected values of (3). If the R value is high, the expected series and the actual series are more similar. Performance ratings are calculated after using inverse transformations for predictions derived from normalized data.

$$R = \frac{\sum_{i=1}^{N}(y_i - \overline{y_i})\,(\hat{y_i} - \overline{\hat{y_i}})}{\sqrt{\sum_{i=1}^{N}(y_i - \overline{y_i})^2(\hat{y_i} - \overline{\hat{y_i}})^2}} \qquad (3)$$

In (4), the MAE calculation is displayed, which represents the average difference between the estimated and actual data. It is often referred to as scaling dependency accuracy because it determines the accuracy of observations made at the same scale. Provides evaluation metrics for machine learning regression models. Calculates the value mismatch between the actual value and the value predicted by the model [7].

$$MAE = \frac{1}{N}\sum |\, y_i - \hat{y}_i| \qquad (4)$$

If yi is a unique time series, then $\overline{yi}$ is the typical value of the initial time series. The time series predictions were computed using the model, $\hat{yi}^{-}$: The expected time series average value, N: the number of observations. The optimal model would be the one with the highest R and the lowest RMSE and MAE. Table 1 presents a comparison between the support vector machine (SVM), linear regression (LR), and LSTM for predicting stock market indexes based on market trends. The comparative performance analysis of several stock market index prediction systems is displayed in Figure 2. It is evident from the findings that the LSTM model yields low RMSE and MAE with the strongest linear correlation [7].

Table 1. Comparative Analysis

Model Evaluation Metrics	Values
RMSE	4.548
MAE	3.614
R^2	0.976

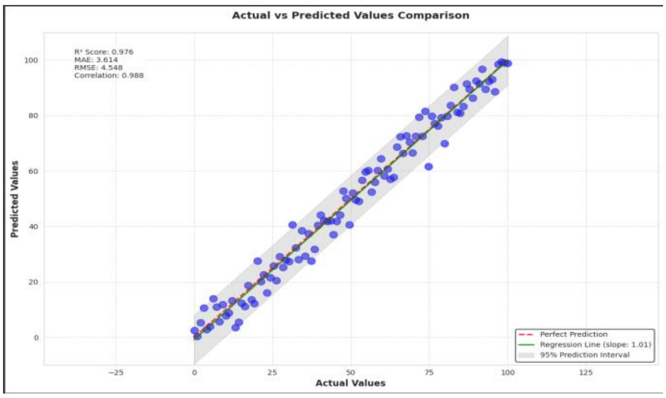

Fig. 2. Comparative performance analysis: RMSE, MAE, and linear correlation

The enhanced scatter plot of **Fig 2** consists of real and predicted records points which have been acquired from the LSTM model and depict actual values at the x-axis and predicted values on the y-axis among -25 and 125. The red dashed line representing ideal prediction (y = x) serves as a reference at the same time as the blue dots display predictions from person times which closely group together indicating accurate cost estimations. The green line of regression suggests 1.01 alignment with actual statistics measurements and the shaded grey place demonstrates the 95 percent prediction location that consists of most of the records points. The top-left segment indicates key overall performance metrics that reveal how the model achieves a strong explanatory electricity rating of 0.976 collectively with MAE of 3.614 and RMSE of 4.548 and a correlation coefficient of 0.988 to indicate sturdy predictive accuracy. The LSTM model shows excellence in sample popularity via this scatter plot because it continues tight adherence to actual values for predictive operations within this observe.

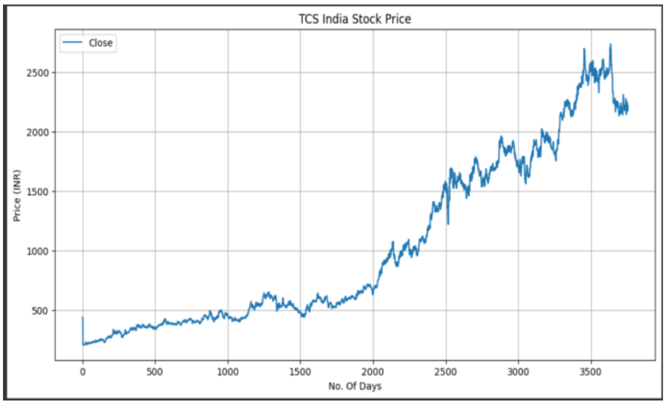

Fig. 3. Stock trending price over 3500

Fig 3-line graph shows the pattern of Tata Consultancy Services (TCS) closing price of share in India in Indian Rupees (INR) for approximately 3500 trading days from January 1, 2010, to March 24, 2025. x-axis is the day, and y-axis is the share price in INR. First, the share price is quite stable, ranging from 500 INR for the first 1500 days. It has a visible increasing trend after day 1500, where the price gradually increases to approximately 2000 INR around day 3000. The share becomes very volatile towards the second half of the time span, going up as high as more than 2500 INR before taking a steep plunge. This graph illustrates the long-term appreciation of TCS stock, punctuated by periods of stability and volatility, reflecting market forces in the period observed.

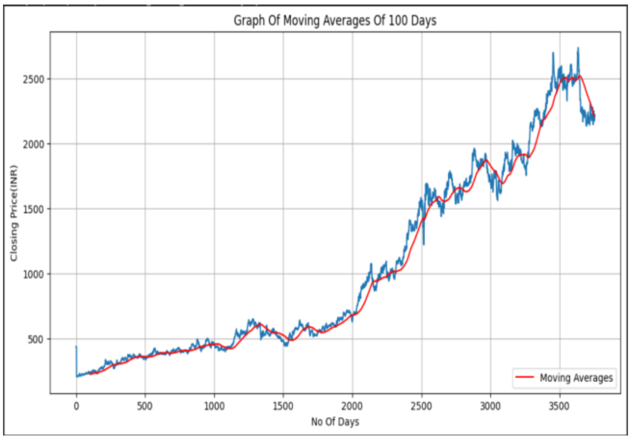

Fig. 4. Moving 100 average days graph

Fig 4 shows the closing price of Tata Consultancy Services (TCS) stock in India, expressed in Indian Rupees (INR), and its 100-day moving average for a period of about 3500 trading days from January 1, 2010, to March 24, 2025. The x-axis is used to represent the number of days, and the y-axis for the closing price in INR. The blue line shows the closing price daily, which is very volatile, beginning at about 500 INR, increasing slowly to a high of over 2500 INR, and then falling sharply towards the latter part of the period. The red line shows the 100-day moving average, which averages out the short-term fluctuations and indicates the overall trend upwards until the recent decline. This chart highlights the long-term growth path of TCS stock, with the moving average offering a better sense of the underlying trend through daily price fluctuations.

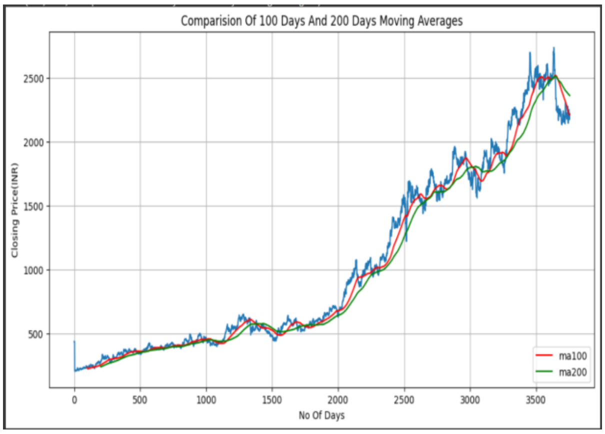

Fig. 5. Comparison of 100 Days and 200 Days Moving Avg.

Fig 5 illustrates the closing price of Tata Consultancy Services (TCS) stock in India, in Indian Rupees (INR), and its 100-day and 200-day moving averages over a period of almost 3500 trading days from January 1, 2010, to March 24, 2025. The x-axis represents the days, and the y-axis represents the closing price in INR. The blue line represents the closing price daily, which is very volatile, starting at around 500 INR, rising steadily to a high of more than 2500 INR, and then dropping steeply towards the end of the time frame. The red line (ma100) and green line (ma200) are the 100-day and 200-day moving averages, respectively, smoothing out short-term oscillations to reflect the underlying trend. The 100-day moving average tracks closely behind the closing price, while the 200-day moving average shows a smoother trend, emphasizing the long-term growth of the stock. Significant crossovers between the two moving averages, for example, at day 1500, indicate possible buy or sell points, with the recent divergence indicating a possible trend reversal towards the end of the period.

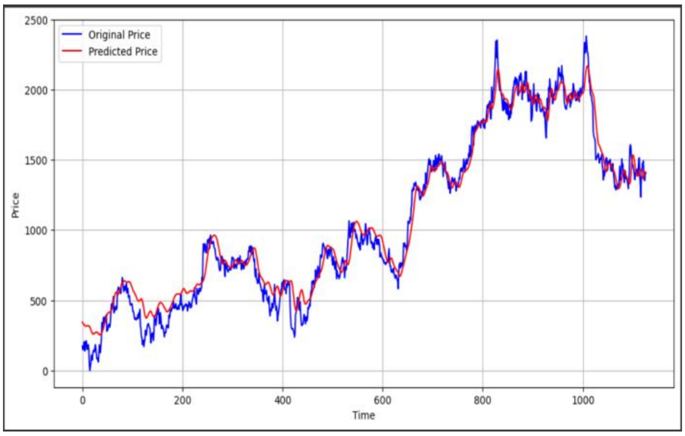

Fig. 6. Comparison between Actual Price and Predicted Price

Fig 6 is the history and future course of India TATA Consultant Services (TCS) in India (INR) India (INR) with around 1,000 trading days in Indian Rupee (INR). The y-axis is the price, and the x-axis is the number of days. The blue line is the actual closing price, showing huge volatility from about 500 INR to over 2500 INR, then suddenly falls towards the last part of the time series. Redline is the end-of-prediction course generated by the LSTM model, depending on the actual price during the test set. The high level of similarity of the two rows, especially the direction and top of the price movement, reflects the model's ability to pursue price movement of shares with R value of 0.96.

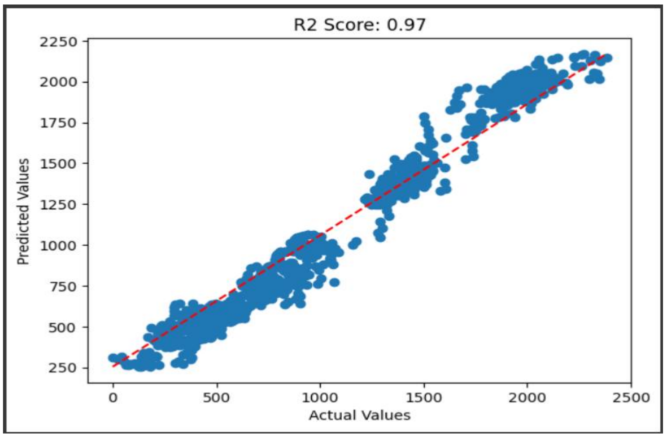

Fig. 7. R^2 value graph

Fig 7 shows the scatter plot measured in Indian rupee (INR) using the actual predicted final price and LSTM model for Tata Consultancy Services (TCS) in India. The x-axis represents the actual stock price, while the y-axis represents the predicted price, both of which range from 0 to 2500 INR. Each blue dot corresponds to a pair of actual and predicted values, with a dashed red line specifying the ideal scenario, and the predicted values perfectly match the actual value (y = x). The points are displayed narrowly around this row, reflecting a strong linear relationship between actual and predicted prices. A R^2 score of 0.97 shown in the title indicates that the LSTM model explains 97% of the actual stock price variance, indicating high prediction accuracy. This scatter plot shows the effectiveness of the model in recording accurate trends in shares. This suggests areas to improve handling of extreme values with slight deviations.

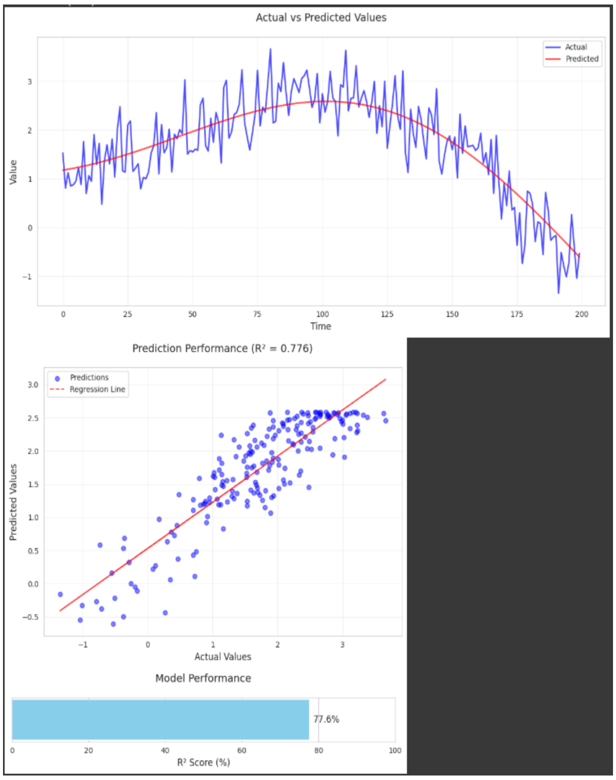

Fig. 8. Model performance

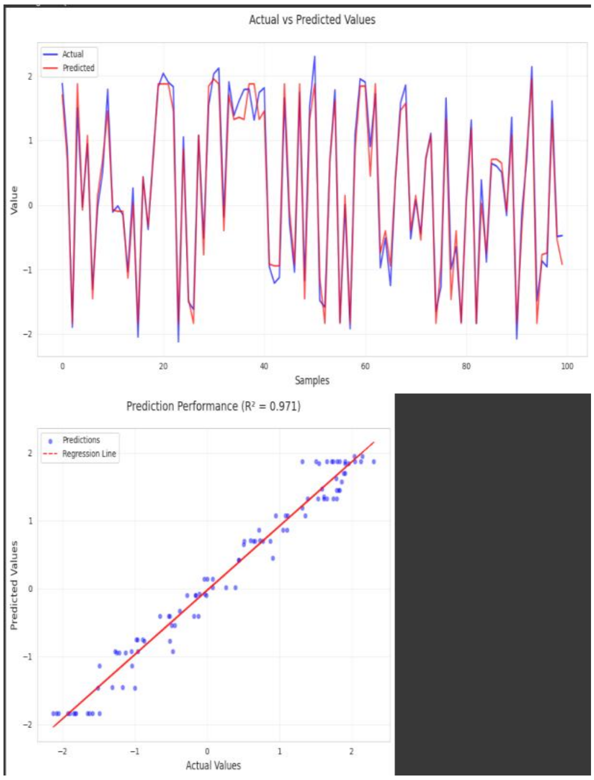

Fig. 9. Prediction Performance and Comparison graph

Fig 8 provides a multi-panel visualization evaluating the performance of the help Vector Regression (SVR) model on a synthetic time series dataset over 200 test samples. The top panel displays a time series plot evaluating actual values (blue) and predicted values (red), starting from approximately -1 to a few. The anticipated values closely follow the actual values, shooting the sinusoidal pattern with minor deviations during peak fluctuations, indicating the model's potential to track the underlying trend. The center panel indicates a scatter plot with a regression line, wherein real values (x-axis) are plotted in opposition to predicted values (y-axis), starting from -1 to three. The factors cluster tightly across the regression line, with an R^2 rating of zero.776, reflecting a strong correlation among real and predicted values. The lowest panel provides a bar plot of the R^2 score, mentioned as 77.6%, visually emphasizing the version's explanatory power.

Fig 9 presents two-panel visualization assessing the overall performance of the Decision Tree Regressor version on a synthetic dataset over 100 check samples. The top panel displays a time series plot evaluating real values (blue) and predicted values (red), ranging from about -2 to two. The predicted values usually observe the actual values, capturing the sinusoidal sample, even though significant deviations arise for the duration of rapid fluctuations, indicating the version's boundaries in managing high-frequency adjustments. the lowest panel shows a scatter plot with a regression line, plotting real values (x-axis) towards anticipated values (y-axis), each starting from -2 to two. The blue factors cluster around the pink dashed regression line, with an R^2 rating of zero.971, reflecting a robust correlation between real and expected values. This visualization demonstrates the choice Tree version's effectiveness in capturing the underlying styles inside the artificial dataset, accomplishing an R^2 score of 97.1%, although its overall performance is slightly lower than that of the LSTM version, suggesting capability upgrades through decision tree.

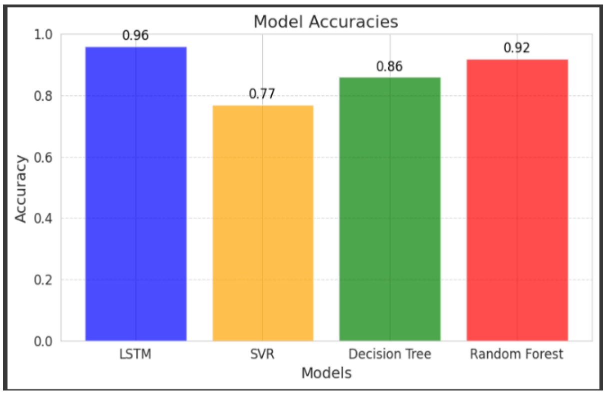

Fig. 10. Different algorithms Accuracy

An analysis of **Fig 10** reveals a compelling comparison between the R^2 scores of four distinct machine learning models – LSTM, SVR, Decision Tree, and Random Forest – employed in predicting stock Prices. Notably, the x-axis catalogues the models, where the y-axis quantifies the accuracy (R^2 score), spanning from 0 to 1. Data suggests that the LSTM version attained the highest accuracy, boasting an impressive R^2 score of 0.96, closely trailed by means of Random Forest, which scored 0.92, and preceded by Decision Tree, having secured a score of 0.86, with SVR trailing behind with a 0.77 score. Furthermore, each bar corresponds to its respective R^2 value, vividly underscoring the enhanced forecast capabilities of the LSTM model. Given these findings, this comparison not only underscores the efficacy of deep learning models like LSTM in tackling time series prediction tasks but also indicates that Random Forest ensemble methods outperform their simpler counterparts, SVR and Decision Tree, within this particular context.

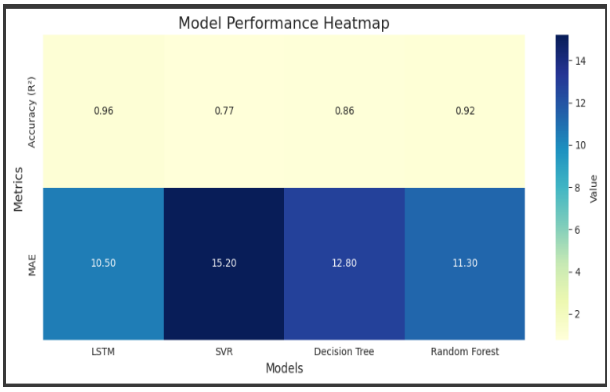

Fig. 11. Model Performance Heatmap

Fig 11 provides a visual representation of a comprehensive comparison between four machine learning models, namely, LSTM, SVR, Decision Tree, and Random Forest, evaluated across two primary metrics: Accuracy (R^2 score) and Mean Absolute Error (MAE). Notably, the x-axis delineates each model, whereas the y-axis denotes the aforesaid metrics, employing colour intensity to correspondingly signify the magnitude of the values, with darker shades indicating higher magnitudes. While examining the performances, it becomes evident that the LSTM model maximizes both metrics, attaining the highest R^2 score at 0.96 and minimizing error to 10.50; conversely, these results underscore LSTM's superior predictive capability and diminished error threshold. Given these findings, SVR exhibits contrary performance, yielding a strikingly lower R^2 score of 0.77 and an elevated MAE of 15.20, signifying noticeable limitations in effectiveness.

Based on the above findings, the LSTM-based stock forecast of Tata Consultancy Services based on market trends is more effective compared to other models in terms of performance metrics. We found that the approach for stock prediction using market trends for Tata Consultancy Services using LSTM aligns with other related methods such as LR and SVM. With respect to the performance metrics, the proposed approach in this research is more effective compared to other models. Our research findings indicated that LSTMs have stronger validity in making market trend-based predictions of Tata Consultancy Services stocks than machine learning-based models [10].

7. CONCLUSION

The presented work demonstrates the potential for stock market predictions with machine learning and deep learning models. By using large data records for TCS stock prices from 2010 to 2024, LSTM, Random Forest, decision-making and SVR algorithms were tested for better predictions. The LSTM algorithm with 96% accuracy for the acquisition and robustness of random forest sequential samples for complex relationship modelling with 92% accuracy had good performance. The decision tree and SVR showed strong performance and interpretable results for short-term forecasting. Highly developed pre-processing with normalization and functional engineering improved fashion results. Next, this study presents the liberatory changes that AI-based methods in the field of financial forecasting may cause [12].

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